

Secretary Problems with Biased Evaluations using Partial Ordinal Information

Jad Salem

Georgia Institute of Technology, Atlanta, GA
jsalem7@gatech.edu

Swati Gupta

Georgia Institute of Technology, Atlanta, GA
swatig@gatech.edu

The k -secretary problem deals with online selection of at most k numerically scored applicants, where selection decisions are immediate upon their arrival and irrevocable, with the goal of maximizing total score. There is, however, wide prevalence of bias in evaluations of applicants from different demographic groups (e.g., gender, age, race), and the assumption of an algorithm observing their true score is unreasonable in practice. In this work, we propose the poset secretary problem, where selection decisions must be made by observing a partial order over the candidates. This partial order is constructed to account for uncertainty in applicant scores. We assume that each applicant has a fixed score, which is not visible to the algorithm and is consistent with the partial order.

Using a random partitioning technique from the matroid secretary literature, we provide order-optimal competitive algorithms for the poset secretary problem and provide matching lower bounds. Further, we develop the theory of thresholding in posets to provide a tight, adaptive-thresholding algorithm under regimes where k grows quickly enough, thus matching the adaptiveness shown for the classical k -secretary problem. We then study a special case in which applicants belong to g disjoint demographic groups and the bias is group-specific. We provide competitive algorithms for adversarial and stochastic variants of this special case, including a framework, GAP, for parallelizing any vanilla k -secretary algorithm to the group setting. Finally, we perform a case study on real-world data to demonstrate the responsiveness to data and the impact of our algorithms.

Key words: secretary problem, bias, online selection, affirmative action

1. Introduction

With the rise of big data and the proliferation of machine learning in important decision-making processes, automated résumé-screening and selection algorithms have become commonplace (Raghavan et al. 2020, Sánchez-Monedero et al. 2020). Hiring decisions are often made based on numerical evaluations of applicants, such as grades, standardized test scores, number of publications, ratings by selection committees, etc., and companies have emerged to assist with automating these decisions (e.g., Indeed, AspiringMinds (2020)). Although scoring applicants provides a numeric scale on which to compare them, it is often unclear what the impact of their training or

their socio-economic background is on test scores. Studies have shown that experiences with racial and gender stereotyping can have adverse effects on test performance (see, e.g., Spencer et al. (1999), Steele and Aronson (1995)). Further, economic status differences and the differences in perceived rewards and costs of the tests can also produce differences in ability test scores between high-income and low-income individuals (Lovaglia et al. 1998). Even if performance is equal, applicants may be penalized due to implicit bias of evaluators (Moss-Racusin et al. 2012), and simply scrubbing protected information can be ineffective in mitigating such biases (De-Arteaga et al. 2019, Dwork et al. 2011). This further becomes a problem in practice as automated screening methods that use real-world data pick up on biased trends in the underlying data. For example, Amazon experimented with such a résumé-filtering algorithm, but scrapped the project when evidence of algorithmic sexism¹ emerged (Goodman 2018).

Many measures for counteracting bias have been applied in practice; we refer the reader to an extensive survey by Raghavan et al. (2019) on industry standards for addressing bias. In particular, seven out of the 18 vendors of algorithmic screening tools that the authors considered publicly discussed “compliance or disparate impact” on their websites. Three of the vendors specifically mentioned adherence to the $4/5$ rule, which requires that the selection rate among any protected group be at least $4/5$ that of any other group. This is typically applied as a safeguard against discrimination lawsuits, but such quota systems are coarse and arguably non-individualistic. Moreover, such quotas may simply set up applicants for failure unless their evaluations by hiring committees are changed fundamentally to account for structural and systemic biases (Sanchez 2022).

Moreover, there is growing literature in recent years to model and address such bias-related issues behaviorally (Cavicchia 2015), mathematically (Corbett-Davies et al. 2017, Dwork et al. 2011, Pierson et al. 2019), and by studying societal systems (Lum 2016, Larson et al. 2016). However, little is known about the impact of bias and the effectiveness of these measures when dealing with implicit bias in data. Kleinberg and Raghavan were the first to show that in offline selection, the Rooney Rule can in fact increase the scores of selected applicants (under certain parameters of their model). Our attempt is to *provide a data-driven model of bias, and to answer a similar question for online selection*.

In this work, we treat bias as a form of uncertainty in applicant evaluations, preventing an accurate ranking of the applicant pool. We propose to forego a total ranking of applicants and make decisions using a partial order² on the set of applicants, which serves as a counterpoint to the

¹ This might have been due to many reasons, e.g. the data that was fed into Amazon’s algorithm might have already incorporated existing biases of the human hiring committees.

² One can turn to potential sources of inaccurate evaluations (e.g., in a study by Steele and Aronson (1995), women’s performance on a math test declined when told that the test produced gender differences, likely pointing to stereotype threat as a cause) in the construction of a partial order.

common method of ranking applicants for the purpose of making selections. Our approach has the potential to address bias from a broader lens that lends itself to being compatible with U.S. caselaw regarding fair hiring practices (Salem et al. 2022). The use of total orderings over applicants to make selections is a design choice which must be validated to be legally viable. However, precedent in U.S. caselaw indicates that reducing applicants to numbers is generally disfavored in the hiring context. Instead, it supports actively accounting for uncertainty and making decisions based on a holistic review of the applicants along multiple evaluation criteria.

Given a batch of applicants and a partial order (encoding uncertainties or biases) over them, it is easy to make offline decisions by (randomly) selecting the highly ranked applicants in the poset. However, at multiple junctures in the hiring pipeline, the problem is not of a completely offline nature. There is an online aspect to the selection process, given the competitive nature of the market and high volume of applicants. In this work, apart from proposing partially ordered sets to counteract bias in evaluations data, we focus on developing secretary algorithms that are competitive given partial ordinal information. Our work is the first to consider this adaptation of the secretary problem for multiple selections. We highlight our contributions next:

1. **Poset Model and Fairness.** We propose the *poset model of bias*, which allows the algorithm access to a partial order (encoding uncertainties and biases) on the set of applicants consistent with the total order induced by true scores (Sec. 3). We show how this model generalizes several natural notions of bias such as the group model of bias (Kleinberg and Raghavan 2018). This is the first paper to use the construction of partial orders as a bias mitigation strategy. Next, we propose *ordinal fairness* (OF) which expands on the notion of conditional demographic parity, incorporating the partial ranking of applicants and enforcing monotonicity in selection rates with increasing rank. We treat OF as a hard constraint in the design of algorithms.
2. **Poset Secretary Algorithms for Small k .** We propose the poset secretary problem, where the goal is to select k applicants with the highest *true* scores from a pool of N , while the algorithm has access only to the partial order on the applications. We claim that bias-agnostic algorithms are not competitive under this setting (Sec. 4), and that any algorithm for the poset secretary problem must incur a competitive ratio of at least ω , where ω is the width³ of the poset (Section 5.1). We discuss challenges to developing poset secretary algorithms in Section 5.2, and we provide an algorithm in Section 5.3 requiring an estimate of the width as input. Next, in Section 5.4, we provide an order-optimal $\mathcal{O}(\omega)$ -competitive algorithm based on the random partitioning technique from the matroid secretary literature (Babaioff et al. 2009) which does not require a width estimate. The main challenge to this approach is learning

³ The width of a poset is the maximum number of mutually incomparable elements.

how “informative” the partial ranking is, which is important for deciding how extensively to sample. We address this by estimating the width of the poset from an initial sample.

3. **An Asymptotically Tight Poset Secretary Algorithm.** Though the above algorithms are order-optimal and implementable in practice due to their explainability, the competitive ratio does not diminish as the number of selections increases, contrary to results in the current secretary literature (Kleinberg 2005). To address this, in Section 5.5, we provide a novel notion of thresholding in partially ordered sets, so that applicants with relatively high potential (according to the partial order) meet the threshold. Our algorithm periodically updates the threshold, thus extending Kleinberg’s adaptive thresholding technique (Kleinberg 2005). The resulting algorithm achieves a tight competitive ratio of $\omega \cdot (1 + o(1))$ as $k \rightarrow \infty$ under the regime where $(\log N)/\sqrt{k} \rightarrow 0$.
4. **Algorithms for the Special Case of Group Bias.** A special case of the poset bias model is the group bias model, which assumes that applicants belong to g disjoint groups, and bias is consistent within groups. This lends a total order on applicants within the same group, and no comparisons on applicants across different groups. In Sec. 6, we simplify our poset algorithms to this setting, resulting in an $\mathcal{O}(g)$ competitive ratio, and we obtain an improved competitive ratio of $\mathcal{O}(1)$ when the group membership and scores are assigned randomly (see Table 1). Moreover, we show that typical vanilla secretary algorithms can be parallelized across the groups in a way that increases their competitive ratio by only a factor of $\mathcal{O}(g)$. This gives a recipe to convert any algorithm in the vanilla secretary setting, to the group-bias setting⁴.
5. **Computational Experiments.** In Section 7, we test our algorithms on the AMEO 2015 dataset,⁵ which includes gender, GPA, college tier, and computer programming test scores for college graduates, among other features. We train a linear regression model to predict computer programming scores for three different shifted distributions of scores, and use gender-specific errors in the resulting predictions to construct a partial order over applicants. We compare selection rates of four algorithms across gender in the secretary setting and observe that enforcing uniform quotas is insensitive to distributional changes in the data, whereas the poset-based algorithms follow the distribution of scores in the data better. Conversely, the poset-based methods tracked changes in score distributions less than vanilla algorithms, since they give applicants more benefit of the doubt by design.

We believe that our work has important practical implications, specifically with regard to understanding best practices for applicant-screening with limited information. Our random partitioning

⁴ Typically parallelizing over posets requires some reduction to the chains of a poset, which is challenging in an online setting.

⁵ The dataset is available at <http://research.aspiringminds.com/resources/#datasets>.

	Competitive ratio	Algorithm		Competitive ratio	Algorithm
Poset model			Group model		
ω known	$(\omega + 1)e^2$ (Corollary 5.1)	PROXYLABEL (Algorithm 1)	Adversarial	$(g + 1)e^2$ (Corollary 5.1)	PROXYLABEL (Algorithm 1)
ω unknown	$e^3(4\omega + 2)(1 + o(1))$ (Proposition 5.3)	POSETLABEL (Algorithm 2)	Adversarial	$gf(k/g)$ (Proposition 6.1)	GAP (Algorithm 4)
ω unknown $\omega \leq \log k$	$\omega \left(1 - \frac{38 \log N}{\sqrt{k}}\right)^{-1}$ (Corollary 5.2)	ADATHRESHOLD (Algorithm 3)	Stochastic	$2e(1 + o(1))$ (Proposition 6.2)	GAPCAP (Algorithm 5)

Table 1 Summary of poset secretary algorithms for selecting k out of N applicants. In the poset model of bias, the algorithm has access to partial ordinal rankings of applicants, and the competitive ratio is given in terms of the width ω of the poset. In the group model of bias, the number of groups g is known to the algorithm; in the stochastic setting, both group membership and scores are random. For Gap, the algorithm being parallelized has competitive ratio $f(k)$ for k selections. All of these algorithms satisfy ordinal fairness.

algorithm can be adapted to practice by forming multiple independent review committees to which applicants are randomly assigned and limiting the number of selections each committee can make. Our theory of thresholding in posets can be used to select applicants based on periodically updated thresholds, which can be implemented in practice. Adaptive thresholds simply represent improved estimates of the applicant pool as more applicants are seen. Our work shows that these techniques give order-optimal algorithms for the poset secretary problem.

2. Related Work

Secretary Problems: Our work adds to the expansive literature on secretary problems by relaxing the assumption that the algorithm has access to true scores of applicants. The classical secretary problem is to select in online fashion at most one element from a randomly ordered weighted set of known size. The classical algorithm for the problem sets a threshold based on a sample of the elements and selects the first non-sampled element exceeding the threshold (Dynkin 1963).

We consider the k -secretary problem, where up to k elements can be selected. There are two techniques which commonly appear in the secretary literature to handle multiple selections: *adaptive thresholding*, in which the threshold is periodically updated, and *random partitioning*, in which elements are randomly partitioned into multiple sets, and selection capacity is distributed across the sets. Our two main algorithms, POSETLABEL and ADATHRESHOLD, use these techniques, respectively. Adaptive thresholding was used by Babaioff et al. (2007) to generalize the classical secretary algorithm while maintaining its e -competitiveness; it was also notably used by Kleinberg (2005) to obtain a tight (as $k \rightarrow \infty$) competitive ratio of $(1 - \frac{5}{\sqrt{k}})^{-1}$. The random partitioning technique has been commonly used in matroid secretary problems (Babaioff et al. 2009, Soto 2013), which we discuss below.

Many other interesting constraints have been studied in the secretary literature, for example, knapsack constraints: each element has both a score and a weight, and the total weight of selected

elements cannot exceed some number (Babaioff et al. 2007). In another line of work, the elements are assumed to form the ground set of some matroid, and selected elements are required to form an independent set. While an $\mathcal{O}(\log \log \text{rank})$ -competitive algorithm was given in Feldman et al. (2015), it remains open whether this problem admits a constant-competitive algorithm. When weights are randomly assigned to the elements, however, one can exploit the principal partition of the matroid to achieve a constant competitive ratio (Soto 2013). These problems all assume access to the true rankings of elements, as opposed to our problem, which assumes only a *partial* ranking.

As our work assumes limited information on scores and rankings of applicants, we discuss other work on limited-information secretary problems. Secretary problems in which the algorithm only has access to rankings of applicants, and not numerical scores, are referred to as *ordinal secretary problems*, and several ordinal matroidal variants have been studied (Soto et al. 2018). Some variants limit information even further, allowing for comparisons between only select pairs of applicants. Among such problems is the Q -queue, J -Choice, K -best secretary problem, which assumes applicants are randomly split into Q groups of equal size, and comparisons cannot be made between applicants of different groups. This differs from our setting, in which the partial order is deterministic and not restricted to be a union of disjoint chains.

More generally, some work has been done on secretary problems with partial ordinal information on rankings, with the objective of picking any maximal element (Kumar et al. 2011). There are three differences between the work of Kumar et al. and ours: (1) we allow for multiple selections, whereas their work allows for only one selection. Their techniques do not easily extend to our multiple-choice setting, nor does their lower bound; (2) they assume prior knowledge of the number of elements *and* the number of maximal elements, whereas we only assume access to the number of elements; and (3) their objective is to select any maximal element, whereas ours is to maximize the total score of selections according to some unknown scoring that is consistent with the rankings. Our objective is harder in the sense that routinely selecting the worst (by score) of the maximal elements will yield good performance by their objective and bad performance by ours. We consider the first two differences to be significant, and the third to be minor.

Fairness in Hiring and Selection Problems: Our work adds to the literature on selection under biased information. The works of Kleinberg and Raghavan (2018) and Celis et al. (2020) consider the scenario where candidates come from various demographic groups, and their observed scores are scaled by a constant dependent on their group membership. In contrast, the work of Emelianov et al. (2020) assumes observed scores are unbiased estimates of true scores with group-dependent variance. In both of these cases, the assumed bias corrupts certain comparisons between candidates, and thus a partial order can be constructed to represent the bias. In this way, our methods provide an alternative approach to the quota-based methods discussed in the above papers. The other main

differences between our work and theirs are that (1) our problem is online, and (2) we make no assumption on the underlying bias except that it can be encoded in a partial order.

In terms of fairness constraints, our proposed notion (“ordinal fairness”) is similar in spirit to individual fairness (Dwork et al. 2011), as both enforce consistency in decisions with some underlying structure (a partial order, a metric). More concretely, ordinal fairness is a partial-ordinal analog of conditional demographic parity (Corbett-Davies et al. 2017). Ordinal fairness also bears resemblance to the meritocratic notion of fairness proposed in the UCB context by Joseph et al. (2016). In their work, individuals who are incomparable (in the sense that their respective confidence intervals overlap) are treated equally.

While our work focuses on single-round hiring and mitigating effects of biased evaluations, one may hope that such interventions will lead to long-term reduction of bias in evaluations. Other work has studied long-term effects of affirmative action and conditions under which certain policies can effectively stabilize the system, resulting in an equilibrium which selects a proportional number of candidates from each demographic group (Mouzannar et al. 2019). In a similar vein of research, Coate and Loury have studied the interplay of affirmative action and employer biases, and the conditions under which a temporary application of affirmative action can lead to long-term benefits for disenfranchised groups (Coate and Loury 1993). Our work complements these by providing a mathematical basis for selecting candidates fairly in spite of bias in evaluations.

We next discuss the problem formulation and the poset model of bias.

3. Preliminaries and Problem Formulation

In Section 3.1, we introduce the poset model of bias as a way to represent uncertainties in evaluations and provide examples showing how the model can be used to account for multiple kinds of bias or uncertainty. In Section 3.2, we introduce ordinal fairness as a desirable quality of selection under poset bias. Finally, in Section 3.3, we formally introduce the poset secretary problem.

3.1. Uncertainty and Partial Orders

We model uncertainty in evaluations using a partial order on the set $\mathcal{C}$ of elements,⁶ in which pairs of elements that cannot be ranked against each other with high enough certainty are deemed incomparable. Each element $a \in \mathcal{C}$ has a true (inaccessible) score $w(a) \geq 0$ signifying their ability or utility to the selection algorithm. We assume that these true scores are distinct, so that they induce a total order $(\mathcal{C}, \preceq_t)$ on the elements, which is not known to the algorithm, but rather the set of such scores will be used to compute the worst-case performance benchmark. The partial order $\mathcal{P} = (\mathcal{C}, \preceq_p)$ which constitutes the feedback must be consistent with the total order; i.e.,

$$a \prec_p b \implies a \prec_t b \iff w(a) < w(b).$$

⁶ For the sake of congruency with the secretary literature, we henceforth refer to applicants as elements.

We next provide examples illustrating how uncertainties can be translated to partial orders.

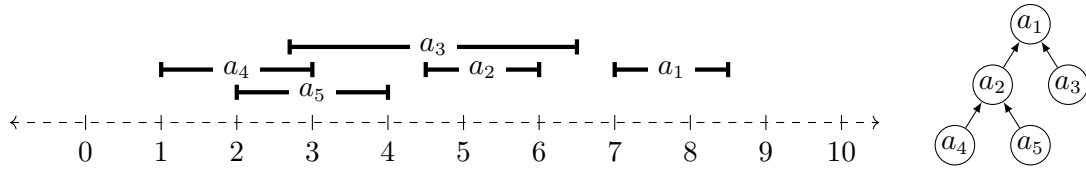
Example 3.1 (Group bias). A special case of our model is the *group bias* model of Kleinberg and Raghavan (2018), which assumes that elements belong to disjoint⁷ demographic groups $G_1, \dots, G_g$, and each group G_i has an associated bias factor $\beta_i \geq 1$. The observed (biased) score of an element $a \in G_i$ is $\tilde{w}(a) = w(a)/\beta_i$. This induces a natural partial order where for any pair of elements $a_1, a_2 \in G_i$, $a_1 \prec a_2$ if $\tilde{w}(a_1) < \tilde{w}(a_2)$. Pairs of elements in different groups are considered incomparable.

Example 3.2 (Discrete evaluations). Suppose each element a has an attribute $x_a \in \{1, \dots, 10\}$, and that this attribute is unbiased and predictive of utility. Then we can define a partial order where $a \prec b$ if and only if $x_a < x_b$. Note that elements with the same attribute value are incomparable.

Example 3.3 (Multiple evaluations). Suppose five elements, $a_1, \dots, a_5$, are evaluated by three metrics or by three committee members. The scores assigned to the elements, respectively, are

$$\begin{bmatrix} 7 \\ 8 \\ 8.5 \end{bmatrix}, \quad \begin{bmatrix} 4.5 \\ 5.5 \\ 6 \end{bmatrix}, \quad \begin{bmatrix} 2.7 \\ 4 \\ 6.5 \end{bmatrix}, \quad \begin{bmatrix} 1 \\ 3 \\ 3 \end{bmatrix}, \quad \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}.$$

As the scores of a_1 were between 7 and 8.5, it may seem likely that the “true” score of a_1 is in the interval $[7, 8.5]$. In this way, these scores naturally induce the following intervals and rankings:⁸



Note that the poset derived in Example 3.3 was constructed from a set of intervals, where comparable elements corresponded to disjoint intervals. Partial orders constructed in this manner constitute an important class called *interval orders*:

Definition 3.1 (interval order) An interval order is a set $\{a_1, \dots, a_N\}$ along with a binary relation $\prec$ that satisfies the following condition: there exists a correspondence between each element a_i and an interval $[\ell_{a_i}, u_{a_i}] \subset \mathbb{R}$, such that $a_i \prec a_j$ if and only if $u_{a_i} < \ell_{a_j}$.

Another way that interval orders can arise is through estimations of bias factors:

⁷ If, for example, the groups G_1 and G_2 intersect nontrivially, then we can simply consider the disjoint groups $G_1 \setminus G_2, G_2 \setminus G_1$, and $G_1 \cap G_2$. This approach non-ideal if there are many demographics being considered, as the number of intersectional groups is exponential in the number of demographics.

⁸ Suppose one wished to select three elements from this poset. Certainly a_1 should be selected, since it is the maximum element. Selection of the other two elements should involve some amount of hedging, since it is unclear which elements are the top three. That said, one might deterministically select a_2 as well, since a_2 is among the top three elements in every linear extension of the poset, in which case only the third selection would involve hedging.

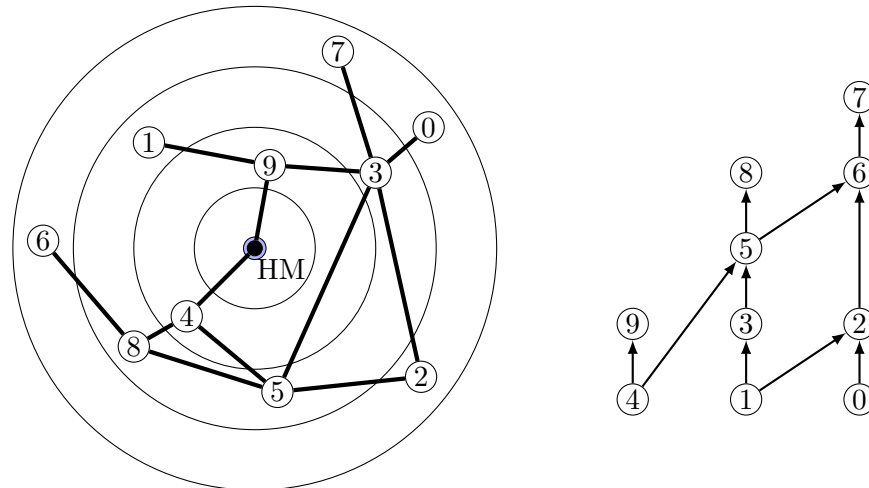
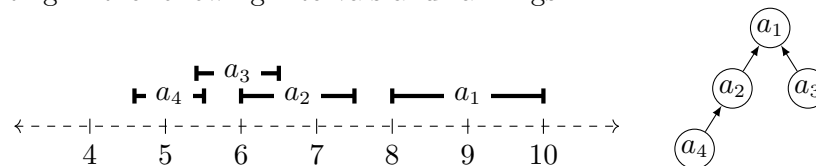


Figure 1 Example of a partial ranking derived from an instance of network bias.

Example 3.4 (Bias factor-derived intervals). Suppose the element pool consists of two groups, with $a_1, a_2 \in G_1$ and $a_3, a_4 \in G_2$. Suppose we know that for any $e \in G_1$, $\beta_e \in [1, 1.25]$, and for any $e \in G_2$, $\beta_e \in [1.25, 1.5]$. If the algorithm observes scores $\tilde{w}(a_1) = 8$, $\tilde{w}(a_2) = 6$, $\tilde{w}(a_3) = 4.33$, and $\tilde{w}(a_4) = 3.67$, then it can be inferred that $w(a_1) \in [8, 10]$, $w(a_2) \in [6, 7.5]$, $w(a_3) \in [5.41, 6.5]$, $w(a_4) \in [4.59, 5.51]$, resulting in the following intervals and rankings:



Notice that adopting the group model of bias here would result in a possibly erroneous ranking of a_3 and a_4 , as the group model does not allow for variation in bias factors within groups.

Example 3.5 (network bias). To account for bias stemming from proximity to the employer, we propose the *network bias* model, in which the hiring manager (HM) and elements (applicants) are vertices in a graph (see Figure 1). The bias that element a experiences depends on the distance⁹ $d(a, \text{HM})$ between the element and the hiring manager in the graph: the greater the distance, the greater the bias experienced. For example, the observed score of an element a can be $\tilde{w}(a) = w(a)/\beta(d(a, \text{HM}))$, where $\beta(\cdot)$ is an increasing function of the distance $d(a, \text{HM})$. In this case, elements at the same distance from HM are comparable (i.e., they form a chain).

Note that the class of posets derived from such networks is different from the class of posets derived from group bias (cf. Example 3.1). The latter consists of posets that are *disjoint* unions of

⁹ i.e., the length of the shortest path.

chains; the former, however, admits some additional comparisons between these chains. Network bias posets and group bias posets are further not subsumed by the class of interval orders.¹⁰

We next discuss fairness in the context of selection from a poset.

3.2. Fairness in Selection from Posets

Equal opportunity for equally qualified applicants is a natural goal to strive for and is often formalized as *conditional demographic parity* (CDP): given a predictive attribute X , group membership G , and decision Y , $\mathbb{P}(Y = 1 \mid X = x, G = g) = \mathbb{P}(Y = 1 \mid X = x)$ (Corbett-Davies et al. 2017).

If one had access to an unbiased and predictive attribute of applicants (e.g., their true scores), then it might make sense to impose CDP: two applicants with the same attribute value must have the same probability of selection. This fairness constraint ensures that irrelevant attributes are not informing or biasing decisions. One drawback of conditional demographic parity is its reliance on an unbiased attribute, which may or may not be available to the decision-maker.

In the case of the poset model of bias, given a partially ordered set of applicants, one can analogously try to treat equally *ranked* or *positioned* applicants *in the poset* equally, in which case conditional demographic parity would be satisfied with respect to their *positions* in the poset. For example, consider a partial order over applicants $\{a, b, c\}$ in which the only comparisons are $b \prec a$ and $c \prec a$ (so, b and c are incomparable). Then, the probability of selection of b should be the same as that of c . In this way, given a partial ranking of applicants, we can apply this poset-variant of conditional demographic parity as an approximation of equal opportunity. To formalize this, we must first define what it means for two elements to be “equally ranked” in a poset. We do so using the notion of an order isomorphism.

Definition 3.2 (order isomorphism) Let $\mathcal{P}_1 = (X, \prec_1)$ and $\mathcal{P}_2 = (Y, \prec_2)$ be partially ordered sets. A bijection $\varphi : X \rightarrow Y$ is an order isomorphism if for all $x_1, x_2 \in X$, we have

$$x_1 \preceq_1 x_2 \iff \varphi(x_1) \preceq_2 \varphi(x_2) .$$

Any two order-isomorphic posets are structurally the same—that is, they have the same number of elements, the same number of minimal and maximal elements, the same set of totally ordered subsets, the same width, etc. An order isomorphism captures this equivalence by mapping minimal elements to minimal elements, totally ordered subsets to totally ordered subsets, and so on. So, if element a can be mapped to element b under an order isomorphism, then the two elements can be considered equally ranked. We can now define ordinal fairness, which is our interpretation of equal opportunity under partial ordinal information.

¹⁰ For example, the poset $\{a, b, c, d\}$ with relations $a \prec b$ and $c \prec d$ (only) cannot be induced by intervals, but can be induced by an instance of network bias or an instance of group bias.

Definition 3.3 (ordinal fairness) We say that a selection algorithm satisfies ordinal fairness (OF) with respect to a partial order if the following conditions hold:

1. (**order parity**) for any order isomorphism φ and any element a , $\mathbb{P}(a \text{ is selected}) = \mathbb{P}(\varphi(a) \text{ is selected})$; and
2. (**monotonicity**) whenever $a \prec b$, $\mathbb{P}(a \text{ is selected}) \leq \mathbb{P}(b \text{ is selected})$.

The first condition is an analog of conditional demographic parity in the partial ordinal setting; in fact, when it is applied to the poset in Example 3.2, we recover the classical notion of conditional demographic parity (see Appendix B); more generally, order parity enforces CDP with respect to the isomorphism classes of the poset. The second condition requires that decisions be more favorable to higher ranking elements. As with conditional demographic parity, OF does not impose group-specific quotas. Indeed, an algorithm satisfying OF may routinely underselect from a given group if that group tends to be low-ranked in the poset. For instance, if there are two groups, G_1 and G_2 , each of size 10, with all members of G_1 ranked higher than all members of G_2 , then an algorithm which selects the top 10 candidates (all from G_1) would satisfy OF despite violating demographic parity.

The fairness of OF comes from the construction of the partial ranking, just as the fairness of individual fairness comes from the construction of the metric. OF (resp. individual fairness) simply ensures adherence to the partial order (resp. metric) which is designed to capture some notion of similarity between applicants. Constraining a selection algorithm to satisfy OF will ensure that the algorithm makes decisions based on the partial ranking, thereby benefiting from any fairness properties the partial ranking might confer.

3.3. Poset Secretary Problem

The poset secretary problem is defined as follows: given a set $\mathcal{C}$ of N elements, there is an underlying poset $\mathcal{P} = (\mathcal{C}, \preceq)$ such that $w(a) < w(b)$ whenever $a \prec b$. The goal of the poset secretary problem is to make irrevocable online selection decisions as elements arrive (in a uniform random arrival order), given access to N and a selection capacity k . Our assumed arrival and decision structure is typical of secretary problems¹¹ except for differences in the information that is observed. At any point in time, the algorithm can observe rankings according to $\mathcal{P}$ on the elements that have arrived so far. The objective is to minimize the competitive ratio subject to ordinal fairness, where competitive ratio is defined as follows:

¹¹ In fact, a constant competitive ratio cannot be achieved if N is unknown (Gharan and Vondrák 2011).

Definition 3.4 An online algorithm¹² $\mathcal{A}$ is α -competitive¹³ if the expected performance ratio is at least $\frac{1}{\alpha}$. In other words, it is α -competitive if for any instance $I = (w, \mathcal{P})$,

$$\rho(I) := \mathbb{E} \left[\frac{w(\text{Alg}(I))}{\text{OPT}(I)} \right] \geq \frac{1}{\alpha},$$

where $\text{OPT}(I)$ is the offline optimum (i.e., the largest total score which is achievable with knowledge of the true scores of all elements), and $\text{Alg}(I)$ is the output of $\mathcal{A}$. The expectation is taken over the possible orderings of elements and any internal randomness in the algorithm. If there is randomness in the instance generation, then the expectation is taken over that randomness as well; note that in this case, $\text{OPT}(I)$ would be a random variable.

Given the uniform random arrival order of applicants, we claim that satisfying ordinal fairness is quite easy as shown in the next proposition; the more challenging aspect of our problem will be minimizing the competitive ratio.

PROPOSITION 3.1 Let $\mathcal{P}$ be a poset of size N , and suppose an algorithm has the property that $a \prec b$ implies $\mathbb{P}(a \text{ is selected}) \leq \mathbb{P}(b \text{ is selected})$. If the algorithm makes decisions based solely on arrival order and $\mathcal{P}$, then it will satisfy ordinal fairness.

Proof. Let φ be any order isomorphism of $\mathcal{P}$. Let Alg denote the set of selected elements, S_N the set of permutations of N elements, and τ_a the transposition of a and $\varphi(a)$. Then

$$\mathbb{P}(a \in \text{Alg}) = \sum_{\sigma \in S_N} \frac{1}{N!} \mathbb{P}(a \in \text{Alg} \mid \sigma) \stackrel{(*)}{=} \sum_{\sigma \in S_N} \frac{1}{N!} \mathbb{P}(\varphi(a) \in \text{Alg} \mid \sigma \circ \tau) = \mathbb{P}(\varphi(a) \in \text{Alg}),$$

where $(*)$ follows since a and $\varphi(a)$ are order-isomorphic. $\square$

Having defined sufficient conditions for satisfying ordinal fairness, we point an interested reader to Propositions A.2 and E.2 for examples of algorithms which do not satisfy ordinal fairness.

4. Suboptimality of Bias-Agnostic Algorithms

Naturally, algorithms which are given access only to biased scores and which do not attempt to counteract the bias will suffer in terms of performance. To explore this more concretely, we consider the group bias scenario (Ex. 3.1), where elements are partitioned into groups $G_1, \dots, G_g$, and the observed score $\tilde{w}(a)$ of element $a \in G_j$ is $\tilde{w}(a) = w(a)/\beta_j$. In this case, we show that the competitive ratio of any bias-agnostic algorithm must scale with $\max_{j \in [g]} \beta_j$.

¹² In this paper, we use $\mathcal{A}$ to denote a generic algorithm. When convenient, we include the set of elements and k as parameters to $\mathcal{A}$. For example, $\mathcal{A}(G, k)$ refers to some k -secretary algorithm run on the set G of elements.

¹³ This is referred to as *strictly α -competitive* in Borodin and El-Yaniv (1998).

PROPOSITION 4.1 *Let k be fixed. Then any bias-agnostic k -secretary algorithm $\mathcal{A}$ operating with ordinal information (i.e., only relative ranking of elements with respect to observed scores) in the group-bias setting has competitive ratio at least $\max_{j \in [g]} \beta_j / 3k$. Additionally, if $\mathcal{A}$ is α -competitive in the classical (unbiased) setting, then it is $\alpha \max_{j \in [g]} \beta_j$ -competitive under group bias.*

The first part of this proposition states that any bias-agnostic algorithm has a competitive ratio which is linear in $\max_{j \in [g]} \beta_j$ for fixed k . Conversely, the second part shows that the competitive ratio can only increase by a factor of $\max_{j \in [g]} \beta_j$ under group bias. We present the proof next.

Proof of Proposition 4.1. Since $\mathcal{A}$ is an ordinal secretary algorithm, it induces a sequence of probabilities of selection $(p_1, \dots, p_N)$ independent of the observed scores $\tilde{w}$, where p_i is the probability of selection of the i th highest scoring element (according to $\tilde{w}$).

To construct an instance where $\mathcal{A}$ performs poorly, consider $g = 2$ groups with $(\beta_1, \beta_2) = (1, \beta)$, and choose $N \geq \beta k$. Let $|G_1| = N - 1$, $|G_2| = 1$, $w(a) \in (1, 2)$ for all $a \in G_1$, and $w(a) = \beta$ for all $a \in G_2$. Now let $\tilde{w}(e_1) > \tilde{w}(e_2) > \dots > \tilde{w}(e_N)$ be the ordering of elements according to observed scores (note that e_N is the element in G_2). Then, letting Alg denote the output of the algorithm given the observed scores, we have

$$\mathbb{E}[w(\text{Alg})] = \sum_{i=1}^N p_i w(e_i) = \sum_{i=1}^{N-1} p_i w(e_i) + p_N w(e_N) \leq 2k + p_N w(e_N) \leq 2k + (k/N)\beta \leq 3k.$$

Since $\text{OPT}/\beta \geq ((k-1) + \beta)/\beta \geq 1$, this gives us that $\mathbb{E}[w(\text{Alg})] \leq 3k \leq (3k/\beta)\text{OPT}$, as desired.

To prove the second statement, let $\widetilde{\text{OPT}}$ the biased optimum (the sum of the top k $\tilde{w}$ -scores), and again let Alg the output of $\mathcal{A}$. Then, $\mathbb{E}(w(\text{Alg})) \geq \frac{\text{OPT}}{\alpha}$, and since $\text{OPT} \leq \beta \widetilde{\text{OPT}}$, it follows that $\mathbb{E}(w(\text{Alg})) \geq \frac{\text{OPT}}{\alpha} \geq \frac{\text{OPT}}{\alpha\beta}$ for any input. Hence, $\mathcal{A}$ is $\alpha\beta$ -competitive. $\square$

We discuss the performance of bias-agnostic algorithms for the poset secretary problem given interval orders and for uncapacitated selection problems given a stream of random scores in Appendix A. We next show how to achieve order-optimal competitive ratios under poset bias (Section 5) and for the special case of group bias (Section 6) while satisfying ordinal fairness.

5. Selection under the Poset Model

In this section, we develop the theory for the poset secretary problem. We begin by showing a lower bound on competitive ratio (Section 5.1), then we address some strategies and difficulties for algorithm design in Sections 5.2-5.3, and finally, we provide order-optimal and asymptotically tight algorithms in Section 5.4 and Section 5.5, respectively.

5.1. Width-Dependent Lower Bound

Before discussing algorithms for secretary problems with partial ordinal information, it is worth discussing lower bounds. Intuitively, the less information contained in the poset, the less capacity

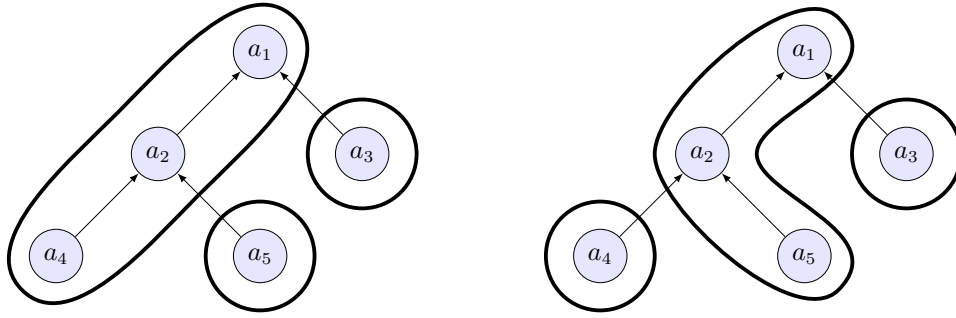


Figure 2 Two chain partitions of a 5-element interval order. An arrow from a_i to a_j means that $a_i \prec a_j$.

we have to make good selections. We quantify this intuition using the width of the poset, which can be thought of as a measure of sparsity.

PROPOSITION 5.1 *Let k and ω be fixed. There is an instance $\mathcal{I} = (\mathcal{P}, w)$ of the poset secretary problem with k selections and width ω such that $\frac{OPT(\mathcal{I})}{\mathbb{E}[w(\text{Alg}(\mathcal{I}))]} \geq \omega$ for any algorithm.*

To justify this, we construct an instance of group bias with ω groups, where most of the score is concentrated in one group. We show that as the amount of total score assigned to the other $\omega - 1$ groups goes to zero, the competitive ratio tends to ω . See Appendix F for the full proof.

5.2. Chain Decomposition

Here we discuss the structure of posets and how it might be exploited in the development of a poset secretary algorithm. In particular, we will attempt to bridge the gap between classical secretary methods and the partial-ordinal setting, and in the process, discuss how these methods can and cannot be extended to account for partial-ordinal information. Since classical secretary algorithms assume a total ordering on elements, the simplest way to extend these algorithms is to run them over totally-ordered subsets (i.e., *chains*) of the poset. As we discuss below, this idea fails to solve our problem, but we carry out the example for pedagogical purposes.

Consider a *chain decomposition* of the poset $\mathcal{P} = (\mathcal{C}, \preceq_p)$; that is, a partition $C_1, \dots, C_\omega$ of $\mathcal{C}$ such that each C_j is a chain and ω is the width of $\mathcal{P}$ (e.g., see Figure 2). Suppose, for the sake of this example, that one had access to chain sizes $|C_1|, \dots, |C_\omega|$ and an oracle which outputs membership in $C_1, \dots, C_\omega$. We will now bound the competitive ratio of the algorithm $\mathcal{A}_p$ which chooses $j \in [\omega]$ uniformly at random and runs an α -competitive algorithm (for some constant α) on C_j . Letting Alg_p be the output of $\mathcal{A}_p$, $\text{Alg}(C_j)$ the output of the α -competitive algorithm on C_j , $\text{OPT}(C_j)$ the total score of the top k elements in C_j , and OPT the overall optimum, we have that

$$\mathbb{E}(w(\text{Alg}_p)) = \frac{1}{\omega} \sum_{j=1}^{\omega} \mathbb{E}(w(\text{Alg}(C_j, k))) \geq \frac{1}{\omega\alpha} \sum_{j=1}^{\omega} \text{OPT}(C_j) \geq \frac{\text{OPT}}{\omega\alpha},$$

thus achieving an order-optimal competitive ratio of $\omega\alpha$.

There are, however, two issues with this approach. First, the structure of the poset is not known in advance, nor are chain sizes and membership oracles. Second, depending on how the chain partition was constructed, the algorithm may not satisfy ordinal fairness¹⁴, since ordinal information is lost in restricting to chains (see Proposition E.2 in Appendix F for a proof of this statement). Moreover, the selection probabilities of elements will depend on the chain partition used (of which there can be multiple, as shown in Figure 2), leaving even more room for unfairness. We will address these issues in the next two subsections¹⁵ by presenting algorithms that do not assume any prior knowledge about the structure of the poset.

5.3. Algorithm Assuming Access to a Width Estimate

As a step toward introducing our main algorithm for the poset setting, we consider the special case where a *width estimate* ω' of the poset is known. Having this (albeit minor) structural knowledge will allow us to develop a competitive algorithm which we will adapt in the next subsection to the general case. The idea behind this width-proxy-based algorithm, which we call PROXYLABEL, is to randomly partition the elements into k sets, collect a random sample, and then select the first maximal element from each of the random sets. Randomly partitioning the elements is one way to “break up” high-scoring elements, thus reducing the competition between the optimal elements. By not restricting to chains, we are considering all available ordinal information, thus avoiding the fairness issues discussed in the previous subsection. Finally, the sampling probability is tuned (using ω') to optimize the competitive ratio.

PROPOSITION 5.2 *Let ω' be the estimated width of the interval order, and let ω be the true width. If $\omega' \geq \frac{1}{c}\omega$, for some $c \geq 1$, then PROXYLABEL (Algorithm 1) has competitive ratio at most $e^c(\omega' + 1)/(1 - \frac{1}{k})^{k-1} \leq e^{c+1}(\omega' + 1)$. Here, we adopt the convention that $0^0 = 1$.*

COROLLARY 5.1 *If the width ω is known, then Algorithm 1 has competitive ratio at most $e(\omega + 1)/(1 - \frac{1}{k})^{k-1} \leq e^2(\omega + 1)$ for any number of selections $k \geq 1$, and this is order-optimal. Here, we adopt the convention that $0^0 = 1$. In particular, if $k = 1$, PROXYLABEL is $e(\omega + 1)$ competitive.*

¹⁴ This approach allowed us to achieve a $3e^2\omega'$ -competitive algorithm for interval orders using Kierstead and Trotter (1981) (see App. E), where ω' is a known upper bound on the width of the poset. However, (i) in general, such an approach may not be possible, since online chain partitioning itself is challenging for general posets, (ii) the competitive ratio incurred additional factors due to the limitations of online chain partitioning, and (iii) this approach requires an upper bound on the width of the poset. We therefore excluded its discussion from this work.

¹⁵ This example additionally highlights a tension between equality of opportunity and equality of outcomes when these chains are demographic groups. Elements in different groups might have the same probability of selection, but all of the selected elements will be from one group. One can think of this as a disparity in *ex-post* (after sampling a group) treatment despite no disparity in *ex-ante* (pre-sampling) treatment. This gives the appearance of discrimination, and certainly does not promote diversity. The algorithms presented in Sections 5 and 6 do not have this issue.

Algorithm 1: PROXYLABEL for poset bias with a width proxy.

input: no. of elements N , estimate ω' on the width (i.e., the width proxy), selection capacity k

- 1 Assign a label $\ell(a) \in [k]$ uniformly at random to each a as they arrive
- 2 Let $m \sim \text{Bin}(N, \frac{\omega'}{\omega'+1})$ and $S \leftarrow \{\text{first } m \text{ elements}\}$ *// sample m elements*
- 3 **for** $a \notin S$ with $\ell(a) = \ell$ **do**
- 4 **if** a is maximal with respect to previously seen ℓ -labeled elements, and no ℓ -labeled element has been selected **then**
- 5 Select a

We next present the proof of the above proposition. The key idea is to first bound the probability that an element a_j among the top k by score is also a maximum element in its assigned label ℓ_j , and to then bound the probability that a maximum-score element in a labeled group is selected by PROXYLABEL. The lower bound for the latter follows from considering an arbitrary chain partition of the ℓ_j -labelled elements: depending on which of the top elements in each chain were sampled, one can bound the selection probability of the maximum-score element.

Proof of Proposition 5.2. Let $w(a_1) > w(a_2) > \dots > w(a_k)$ be the scores of the top k elements. Consider any element a_j among these k elements, and let $\ell(a_j) = \ell_j$. Then the probability that a_j is maximum in its labeled group can be bounded by $1/e$ as follows:

$$\mathbb{P}\left(\{w(a_j) = \max\{w(a) : \ell(a) = \ell_j\}\}\right) = \mathbb{P}\left(\bigcap_{i=1}^{j-1} \{\ell(a_i) \neq \ell_j\}\right) = \left(1 - \frac{1}{k}\right)^{j-1} \geq \left(1 - \frac{1}{k}\right)^{k-1} \geq \frac{1}{e}.$$

We can then bound the probability of selection of a_j by analyzing elements with label ℓ_j . Letting Alg denote the set of elements returned by the algorithm, we have:

$$\mathbb{P}\left(\{a_j \in \text{Alg}\}\right) \geq \mathbb{P}\left(\{a_j \in \text{Alg}\} \cap \{w(a_j) = \max\{w(a) : \ell(a) = \ell_j\}\}\right) \quad (1)$$

$$\begin{aligned} &= \mathbb{P}\left(\{w(a_j) = \max\{w(a) : \ell(a) = \ell_j\}\}\right) \mathbb{P}\left(\{a_j \in \text{Alg}\} \mid \{w(a_j) = \max\{w(a) : \ell(a) = \ell_j\}\}\right) \\ &\geq \left(1 - \frac{1}{k}\right)^{k-1} \underbrace{\mathbb{P}\left(\{a_j \in \text{Alg}\} \mid \{w(a_j) = \max\{w(a) : \ell(a) = \ell_j\}\}\right)}_{=: A} \end{aligned} \quad (2)$$

Note that the inequality in (1) may not be an equality, since a_j can be selected even if it is not a maximum-score ℓ_j -labeled element. Now we bound A . To do so, consider a partition $\{C_1, C_2, \dots, C_{\omega_{\ell_j}}\}$ of the ℓ_j -labeled elements into $\omega_{\ell_j} \leq \omega$ chains, with $a_j \in C_1$. For each chain C_r , let $f_s^{(r)}$ denote the s -th best element in C_r . Then a_j will be selected if (but not only if) the following conditions hold¹⁶:

¹⁶ if $f_2^{(1)}$ does not exist, then we can replace the corresponding event with the empty event. This only increases the chances of selecting a_j .

Algorithm 2: POSETLABEL for partial ordinal information.

input: number of elements N , number to hire k

- 1 Assign a label $\ell(a) \in [k]$ uniformly at random to each a as they arrive
 - 2 Let $m_1 \sim \text{Bin}(N, \frac{1}{2})$ and $S' \leftarrow \{\text{first } m_1 \text{ elements}\}$
 - 3 Let ω' be the width of the subposet S'
 - 4 Let $m_2 \sim \text{Bin}(N, \frac{2\omega'}{2\omega'+1})$
 - 5 Let $S \leftarrow \{\text{first } \max\{m_1, m_2\} \text{ elements}\}$ (so, $S \supseteq S'$)
 - 6 **for** $a \notin S$ with $\ell(a) = \ell$ **do**
 - 7 **if** a is maximal with respect to previously seen ℓ -labeled elements, and no ℓ -labeled element has been selected **then** Select a
-

1. a_j is not sampled (i.e., $a_j \notin S$),
2. the second best element in C_1 is sampled (i.e., $f_2^{(1)} \in S$), and
3. the best elements in all other chains are sampled (i.e., $f_1^{(i)} \in S$ for $i = 2, \dots, \omega_{\ell_j}$).

It follows that

$$A \geq \mathbb{P}(\{a_j \notin S\}) \mathbb{P}(\{f_2^{(1)} \in S\}) \mathbb{P}\left(\bigcap_{i=2}^{\omega_{\ell_j}} \{f_1^{(i)} \in S\}\right) \geq \left(\frac{1}{\omega'+1}\right) \left(1 - \frac{1}{\omega'+1}\right)^\omega. \quad (3)$$

Now suppose that $\frac{1}{c}\omega \leq \omega'$, for some $c \geq 1$. Then, using (2) and (3), we get:

$$\mathbb{P}(\{a_j \in \text{Alg}\}) \geq \left(1 - \frac{1}{k}\right)^{k-1} A \geq \left(1 - \frac{1}{k}\right)^{k-1} \left(\frac{1}{\omega'+1}\right) \left(1 - \frac{1}{\omega'+1}\right)^\omega \geq \left(1 - \frac{1}{k}\right)^{k-1} \frac{1}{e^c(\omega'+1)}.$$

Thus, $\mathbb{E}(w(\text{Alg}))$ can be bounded below by

$$\sum_{i=1}^k \mathbb{P}(\{a_i \in \text{Alg}\}) w(a_i) \geq \sum_{i=1}^k \left(1 - \frac{1}{k}\right)^{k-1} \frac{w(a_i)}{e^c(\omega'+1)} = \left(1 - \frac{1}{k}\right)^{k-1} \frac{\text{OPT}}{e^c(\omega'+1)} \geq \frac{\text{OPT}}{e^{c+1}(\omega'+1)}.$$

□

We have just shown that when a width estimate $\omega' \geq \omega/c$ is known, one can design an $\mathcal{O}(e^c \omega')$ -competitive algorithm for the secretary problem under partial ordinal information. In the next section, we will show how to circumvent this need for a width estimate.

5.4. Width-unknown Algorithm for Poset Secretary

In this section, we present the first poset secretary algorithm, POSETLABEL, which only requires the knowledge of the number of elements as described in our problem formulation (in Section 3). This algorithm is an adaptation of the PROXYLABEL algorithm of the previous section which foregoes the assumption on a width estimate.

Recall that PROXYLABEL uses a width proxy ω' to set a sampling probability. To get around this, POSETLABEL implements a two-stage sampling procedure. First, an initial sample is taken for the purpose of width-estimation. The result is a (potentially inaccurate) estimate of the width, from which we can define a secondary sample size. This two-stage sampling procedure allows the sampling to begin *without knowledge of the ultimate sample size*, which is crucial since the width is a priori unknown. See Appendix D for an example run of this algorithm (Algorithm 2). We next show that this algorithm is $\mathcal{O}(\omega)$ -competitive.

PROPOSITION 5.3 *Under the poset bias setting, Algorithm 2 is $2e^3(2\omega + 1)(1 + o(1))$ -competitive, as the number of elements $N \rightarrow \infty$. Moreover, this algorithm satisfies ordinal fairness, and does not need to know the width of the partial order in advance.*

As was the case with PROXYLABEL, the random partition of elements reduces the probability that the best elements are clustered in one chain. Then, by forming a chain partition of elements using the minimum number of chains, we can bound the probability of selecting a maximal element within its labeled group. Finally, we can show that our estimated width ω' is within a factor of 4 of the true width ω with large probability, giving us the $\mathcal{O}(\omega)$ competitive ratio in the proposition. The full proof follows.

Proof of Proposition 5.3 Let S' be the initial sample of $m_1 \sim \text{Bin}(N, 1/2)$ elements, and let ω' be the width of S' . Let $\text{Alg}_1(\omega')$ denote the output of PROXYLABEL (Algorithm 1) with width proxy ω' , and let Alg_2 denote the output of POSETLABEL (Algorithm 2).

We begin by bounding ω' . Let $A = \{a_1, \dots, a_\omega\}$ be a (maximal) set of mutually incomparable elements, and let $|A \cap S'| = X \sim \text{Bin}(\omega, \frac{1}{2})$. In order to apply our results from Proposition 5.2, we must bound the probability that ω' is close to ω . To that end, letting $X_\omega = \{\omega' \geq \frac{\omega}{4}\}$,

$$\mathbb{P}(X_\omega) \geq \mathbb{P}\left(\left\{X \geq \frac{1}{4}\omega\right\}\right) \geq \frac{1}{2}. \quad (4)$$

After sampling m_1 elements, the algorithm uses its estimated width ω' to sample $m_2 \sim \text{Bin}(N, \frac{2\omega'}{2\omega'+1})$. Now in order to reduce to the analysis of Proposition 5.2, we need to bound the probability that $m_2 \geq m_1$. Letting $Y_N = \{m_2 \geq m_1\}$ (where the subscript N refers to the number of elements), we have that $\mathbb{P}(Y_N) \geq \mathbb{P}(\{m_2 \geq m_1 \geq 1\})$. When $m_1 \geq 1$, we have that $\omega' \geq 1$, and so the probability $\frac{2\omega'}{2\omega'+1}$ of inclusion into S is at least $\frac{2}{3}$. It follows that for $Z_N \sim \text{Bin}(N, \frac{2}{3})$,

$$\mathbb{P}(Y_N) \geq \mathbb{P}(\{m_2 \geq m_1 \geq 1\}) \geq \mathbb{P}(\{Z_N \geq m_1 \geq 1\}) \xrightarrow{N \rightarrow \infty} 1.$$

This is because $\mathbb{E}(m_1) = \frac{N}{2}$, and $\mathbb{E}(Z_N) = \frac{2N}{3}$. Since these expectations differ by $\Theta(\sqrt{N})$ standard deviations, Y_N occurs with high probability, and since Y_N happens with high probability,

$$\mathbb{P}(X_\omega \cap Y_N) \xrightarrow{N \rightarrow \infty} \mathbb{P}(X_\omega). \quad (5)$$

Conditioning on the event $X_\omega \cap Y_N$, it is as if we simply ran Algorithm 1 (PROXYLABEL), and so we can use the analysis from that algorithm.

$$\begin{aligned}
\mathbb{E}(w(\text{Alg}_2)) &\geq \mathbb{P}(X_\omega \cap Y_N) \mathbb{E}\left(w(\text{Alg}_2) \mid X_\omega \cap Y_N\right) \\
&= (1 - o(1)) \mathbb{P}(X_\omega) \mathbb{E}\left(w(\text{Alg}_1(2\omega')) \mid \left\{\omega' \geq \frac{\omega}{4}\right\}\right) && \text{by (5)} \\
&\geq \frac{(1 - o(1))}{2} \mathbb{E}\left(w(\text{Alg}_1(2\omega')) \mid \left\{\omega' \geq \frac{\omega}{4}\right\}\right) && \text{by (4)} \\
&\geq \frac{(1 - o(1))}{2} \left[p \mathbb{E}\left(w(\text{Alg}_1(2\omega')) \mid \left\{2\omega' \geq \omega\right\}\right) \right. \\
&\quad \left. + (1 - p) \mathbb{E}\left(w(\text{Alg}_1(2\omega')) \mid \left\{\frac{\omega}{2} \leq 2\omega' < \omega\right\}\right) \right] && (*) \\
&\geq \frac{(1 - o(1))}{2} \left[p \frac{\text{OPT}}{e^2(2\omega' + 1)} + (1 - p) \frac{\text{OPT}}{e^3(2\omega' + 1)} \right] \geq \frac{(1 - o(1))}{2} \left[\frac{\text{OPT}}{e^3(2\omega' + 1)} \right] \\
&\geq \frac{(1 - o(1))}{2} \left[\frac{\text{OPT}}{e^3(2\omega + 1)} \right],
\end{aligned}$$

where $(*)$ holds for some $p \in [0, 1]$. $\square$

5.5. Adaptive Thresholding: An Improved Algorithm for Large k

While POSETLABEL achieves an order-optimal competitive ratio, the proved bound on competitive ratio is insensitive to N and k . One might expect the achievable performance to improve as k increases, since a small number of bad decisions relative to k may not have a large effect on performance. Indeed, the $(1 - 5k^{-1/2})^{-1}$ -competitive algorithm of Kleinberg (2005) for the vanilla k -secretary problem supports this inclination. In this section, we present and analyze a novel algorithm for the poset k -secretary problem which achieves a competitive ratio of $\omega(1 + o(1))$ as $k \rightarrow \infty$ under the regime where $(\log N)^2 \in o(k)$.

Before presenting the new algorithm, it is worth noting that POSETLABEL is not tight as $k \rightarrow \infty$. To see why, it suffices to consider the vanilla case, where $\omega = 1$. Recall that POSETLABEL randomly partitions the elements into k sets and selects at most one element from each set. We show below that any algorithm with this property fails to achieve a tight competitive ratio (Prop. 5.4).

Definition 5.1 (URP k -secretary algorithm) A uniform-random-partition (URP) k -secretary algorithm is one that (1) assigns each element independently and uniformly at random to one of k sets, and (2) selects at most one element from each set.

PROPOSITION 5.4 Any URP k -secretary algorithm cannot achieve a competitive ratio smaller than $\left(1 - \left(1 - \frac{1}{k}\right)^k\right)^{-1}$. In particular, this means that no URP algorithm can achieve a competitive ratio better than $(1 - 1/e)^{-1}$ asymptotically in k .

Proof. Let Alg^* be the (offline) algorithm that assigns each element to one of k sets uniformly at random, and chooses the best element in each of the sets. Now let $a_1, \dots, a_k$ be the k highest-score elements, with $w(a_1) > \dots > w(a_k)$. Let $\varepsilon_j = w(a_1) - w(a_j)$ for each $j \leq k$, and let $\varepsilon = \sum_{j=1}^k \varepsilon_j$. Then

$$\begin{aligned} \mathbb{E}[w(\text{Alg}^*)] &= \sum_{j=1}^k w(a_j) \mathbb{P}(a_j \in \text{Alg}^*) = \sum_{j=1}^k w(a_j) \left(1 - \frac{1}{k}\right)^{j-1} \\ &\leq w(a_1) \sum_{j=1}^k \left(1 - \frac{1}{k}\right)^{j-1} = w(a_1) \cdot \frac{1 - \left(1 - \frac{1}{k}\right)^k}{1/k} \leq (\text{OPT} + \varepsilon) \left(1 - \left(1 - \frac{1}{k}\right)^k\right). \end{aligned}$$

Letting $\varepsilon \rightarrow 0$, we see that the performance ratio of Alg^* is no better than $\left(1 - \left(1 - \frac{1}{k}\right)^k\right)^{-1}$. Since Alg^* performs better than any URP algorithm, all such algorithms have competitive ratios no better than $\left(1 - \left(1 - \frac{1}{k}\right)^k\right)^{-1}$. Moreover, since $\left(1 - \left(1 - \frac{1}{k}\right)^k\right)^{-1} \rightarrow (1 - 1/e)^{-1}$, we see that URP algorithms cannot achieve a competitive ratio better than $(1 - 1/e)^{-1}$ as $k \rightarrow \infty$. $\square$

Since URP k -secretary algorithms cannot achieve tight competitive ratios, we turn to the *adaptive thresholding* technique which is common in the secretary literature (Kleinberg 2005, Babaioff et al. 2007). In the vanilla setting, this technique involves periodically setting a new threshold, and selecting elements whose scores exceed the most recent threshold. This technique is promising for obtaining a tight competitive ratio since it allows for a small number of poor decisions to be made early on and a larger number of decisions to be made later on with more carefully chosen thresholds. The main challenges of adapting this approach to the poset setting are (1) adapting the notion of a threshold, and (2) bounding the expected scores of elements that meet the most recent threshold.

Let us first define some notions which will be useful in describing the algorithm. In what follows, $\overline{X}$ will denote the complement of a set X .

Definition 5.2 Let $\mathcal{P}$ be a poset. The upset $\mathcal{U}(\tau)$ of a set τ is $\{b : b \succ a \text{ for some } a \in \tau\}$. The downset $\mathcal{D}(\tau)$ of τ is $\{b : b \prec a \text{ for some } a \in \tau\}$. The selection set $\sigma(\tau)$ is the set of elements which are better than or incomparable to elements in τ , i.e., $\sigma(\tau) = \{b \mid b \not\prec a \text{ for all } a \in \tau\} = \overline{\mathcal{D}(\tau)} \setminus \tau$.

In order to achieve a good competitive ratio, we would like as many of the highly ranked elements to be in the selection set $\sigma(\tau)$ of the threshold τ as possible. Since it is unclear from the partial order which elements have the top k scores, the threshold τ must select elements which *could possibly* be among the top k , and not only select elements which are definitely among the top k . This may, for example, involve hedging across the poset: if multiple chains in a given chain partition contain elements that could possibly be in the top k , then it may be necessary for τ to select elements from those chains. To that end, we introduce the notion of an (η, k) -threshold.

Definition 5.3 ((η, k)-threshold) Let $\mathcal{P}$ be a poset of width ω . An (η, k) -threshold $\tau \subseteq \mathcal{P}$ is an antichain with selection set $|\sigma(\tau)| \leq k$ such that the upset is large enough for each element in τ : $|\mathcal{U}(a)| \geq \lfloor (1 - \eta)k/\omega \rfloor$ for all $a \in \tau$.

Any (η, k) -threshold τ hedges across chains containing potentially high-scoring elements, since each element in τ needs to have a large enough upset. The elements that meet such a thresholding antichain τ end up having a high total score for any linear extension of the poset, as shown below.

LEMMA 5.1 Let τ be an (η, k) -threshold of a scored (weighted) poset $\mathcal{P}$, and let OPT denote the sum of the k largest scores in $\mathcal{P}$. Then $w(\sigma(\tau)) \geq OPT \lfloor (1 - \eta)k/\omega \rfloor / k$.

Proof. Let a be among the top $\lfloor (1 - \eta)k/\omega \rfloor$ elements in $\mathcal{P}$ by score. If $a \notin \sigma(\tau)$, then $a \preceq b$ for some $b \in \tau$, in which case

$$\left\lfloor (1 - \eta) \frac{k}{\omega} \right\rfloor > |\mathcal{U}(a)| \geq |\mathcal{U}(b)| \geq \left\lfloor (1 - \eta) \frac{k}{\omega} \right\rfloor.$$

Thus, $a \in \sigma(\tau)$, implying that $w(\sigma(\tau)) \geq OPT \lfloor (1 - \eta)k/\omega \rfloor / k$. $\square$

The above lemma shows that (η, k) -thresholds are effective in selecting high-scoring elements. We next show that such (η, k) -thresholds exist. Together, these two observations indicate that (η, k) -thresholds could be useful in designing algorithms for poset secretary problems.

LEMMA 5.2 (EXISTENCE OF (η, k) -THRESHOLDS) Let $\mathcal{P}$ be a poset of size N and width ω , let $k \leq N$, and let $\eta \in [0, 1)$. There exists an (η, k) -threshold of $\mathcal{P}$.

Proof. Let $(C_1, \dots, C_\omega)$ be a chain decomposition of $\mathcal{P}$, and let $a_{i,1} \succ \dots \succ a_{i,n_i}$ denote the elements of C_i . For each $1 \leq i \leq \omega$, define E_i to be the $\lfloor (1 - \eta)k/\omega \rfloor + 1$ th best element in C_i , if it exists, i.e.,

$$E_i = \begin{cases} \{a_{i, \lfloor (1 - \eta)k/\omega \rfloor + 1}\} & \text{if } \lfloor (1 - \eta)k/\omega \rfloor + 1 \leq n_i, \\ \emptyset & \text{else.} \end{cases}$$

Now let $\tau = \bigcup_{i=1}^{\omega} E_i$. Note that each element $a \in \tau$ satisfies $|\mathcal{U}(a)| \geq \lfloor (1 - \eta)k/\omega \rfloor$, and that $|\sigma(\tau)| \leq \omega \lfloor (1 - \eta)k/\omega \rfloor \leq \lfloor (1 - \eta)k \rfloor \leq k$. However, τ may not be an antichain. In this case, there is some $b_1, b_2 \in \tau$ such that $b_1 \prec b_2$. Now let $\tau' = \tau \setminus \{b_1\}$. We still have the property that $|\mathcal{U}(a)| \geq \lfloor (1 - \eta)k/\omega \rfloor$ for each $a \in \tau'$. Moreover, since $\mathcal{D}(b_1) \subseteq \mathcal{D}(b_2)$, we have that $|\sigma(\tau')| = |\sigma(\tau)| \leq k$. Thus we can repeat this procedure until we arrive at an antichain. $\square$

Now that we have a good notion of thresholding, we can describe our algorithmic approach, which generalizes the vanilla k -secretary algorithm of Kleinberg (2005). The algorithm randomly splits the element stream into two sets, Y and Z , and recurses on Y to select at most $k/2$ elements. Then, a threshold is chosen from Y , and elements in Z who meet the threshold are selected. See Algorithm 3 for a detailed description. We next bound the competitive ratio of this algorithm when an upper bound on the width of the poset is known in advance.

Algorithm 3: ADATHRESHOLD(S, k)

input: slack $\eta = \eta(N, k)$, element stream S of size N , $k \geq 1$

- 1 **if** $k \geq N$ **then**
- 2 Select all elements
- 3 **else if** $k = 1$ **then**
- 4 Run POSETLABEL(S, k)
- 5 **else**
- 6 Sample $m \sim \text{Bin}(N, 1/2)$ and let Y be the first m elements
- 7 Run ADATHRESHOLD($Y, \lfloor k/2 \rfloor$)
- 8 Select any $(\eta(N, k), \lfloor k/2 \rfloor)$ -threshold $\hat{\tau}$ of Y
- 9 While fewer than k selections have been made, select elements in $S \setminus Y$ that meet $\hat{\tau}$

PROPOSITION 5.5 *Let $\mathcal{P}$ be a poset on N elements of width ω , and let $\omega' \geq \omega$ be a known upper bound on ω , and set $D = 4\sqrt{\omega'}$. Then ADATHRESHOLD (Algorithm 3) with $\eta = D\sqrt{\frac{\log N}{k}}$ is $\omega(1 - C\sqrt{\frac{\log N}{k}})^{-1}$ -competitive, where $C = \frac{7+4D}{4-2\sqrt{2}}$.*

Proof. We proceed by induction on k , mimicking the structure of the algorithm. For simplicity, we assume that k is a power of 2. Note that the bound holds trivially if $k \leq D^2 \log N$, since $1 - C\sqrt{\log N/k} \leq 1 - C/D \leq 1 - 4/(4 - 2\sqrt{2}) < 1$, so we assume that $k > D^2 \log N$. Similarly, if $k \geq N$, then we select all elements and the competitive ratio is 1, so the bound trivially holds, since $1 - C\sqrt{\log N/k} \leq \omega$. If $N \geq 2$ and $k \leq 2$, then the bound again becomes trivial:

$$1 - C\sqrt{\frac{\log N}{k}} \leq 1 - C\sqrt{\frac{\log 2}{2}} \leq 1 - \frac{7 + 12\sqrt{2}}{4 - 2\sqrt{2}} \cdot \sqrt{\frac{\log 2}{2}} < 0.$$

Now suppose that $3 \leq k \leq N$, with $k > D^2 \log N$. We bound the performance of the algorithm on Y (the first half, on which we recurse) and $Z = S \setminus Y$ (the second half) separately. Let $\text{Alg}(S)$ denote the set of elements selected by the algorithm over a set S . Letting $\text{OPT}(Y, k/2)$ denote the total score of the top $k/2$ elements in Y ,

$$\begin{aligned}
\mathbb{E}[w(\text{Alg}(Y))] &\geq \left(1 - \frac{\sqrt{2}C\sqrt{\log |Y|}}{\sqrt{k}}\right) \mathbb{E}\left[\text{OPT}\left(Y, \frac{k}{2}\right)\right] / \omega && \text{inductive hypothesis} \\
&\stackrel{(*)}{\geq} \left(1 - \frac{\sqrt{2}C\sqrt{\log |Y|}}{\sqrt{k}}\right) \left(\frac{1}{2} - \frac{1}{4\sqrt{k}}\right) \frac{\text{OPT}}{\omega} \\
&\geq \left(\frac{1}{2} - \frac{1 + 2\sqrt{2}C\sqrt{\log |Y|}}{4\sqrt{k}}\right) \frac{\text{OPT}}{\omega} \\
&\geq \left(\frac{1}{2} - \frac{1 + 2\sqrt{2}C\sqrt{\log N}}{4\sqrt{k}}\right) \frac{\text{OPT}}{\omega},
\end{aligned}$$

where $(*)$ follows directly from a computation of Kleinberg (2005) and is included as Lemma F.1.

We can similarly bound the performance of the algorithm on Z . To that end, note that $\eta < 1$, since $k > D^2 \log N$, so the algorithm is able to choose a threshold $\hat{\tau}$. We next bound the probability that the threshold $\hat{\tau}$ selected by the algorithm is good. Define $\mathcal{B} = \{\tau \in \mathfrak{A}(\mathcal{P}) : |\sigma(\tau)| > k \text{ or } |\mathcal{U}(b)| < (1 - 2\eta)\frac{k}{\omega} \text{ for some } b \in \tau\}$ to be the set of “bad” thresholds (since it selects too many elements or one of its elements has a small upset), where $\mathfrak{A} = \mathfrak{A}(\mathcal{P})$ is the set of antichains of $\mathcal{P}$. Then

$$\mathbb{P}(\hat{\tau} \in \mathcal{B}) \leq \underbrace{\mathbb{P}(|\sigma(\hat{\tau})| > k)}_{=:A} + \underbrace{\mathbb{P}\left(\bigcup_{b \in \hat{\tau}} \left\{|\mathcal{U}(b)| < (1 - 2\eta)\frac{k}{\omega}\right\}\right)}_{=:B}.$$

We bound A and B separately, starting with A . For simplicity, we will assume that $(1 - \eta)k/\omega$ and $(1 - 2\eta)k/\omega$ are integers. Then

$$\begin{aligned} A = \mathbb{P}(|\sigma(\hat{\tau})| > k) &= \sum_{\substack{\tau \in \mathfrak{A} \\ |\sigma(\tau)| > k}} \mathbb{P}(\hat{\tau} = \tau) \leq \sum_{T=k}^N \sum_{\substack{\tau \in \mathfrak{A} \\ |\sigma(\tau)| = T}} \mathbb{P}(\hat{\tau} = \tau) \\ &\leq \sum_{T=k}^N \sum_{\substack{\tau \in \mathfrak{A} \\ |\sigma(\tau)| = T}} \mathbb{P}\left(|\sigma(\tau) \cap Y| \leq (1 - \eta)\frac{k}{2}\right) \\ &\stackrel{(a)}{\leq} \sum_{T=k}^N |\{\tau \in \mathfrak{A} : |\sigma(\tau)| = T\}| e^{-\eta^2 k/2} \\ &\stackrel{(b)}{\leq} \sum_{T=k}^N \binom{T + \omega - 1}{\omega - 1} e^{-\eta^2 k/2} \leq (2N)^{\omega+1} e^{-\eta^2 k/2} =: A', \end{aligned}$$

where (a) follows from Hoeffding’s inequality (since the selection set of τ is a fixed set in the poset, and bounding it’s size in the sample is simply a Binomial) and (b) follows from Lemma F.2 in Appendix F. One can similarly obtain the bound $B \leq N e^{-\eta^2 k/4\omega} =: B'$ by applying Hoeffding’s inequality (see Lemma F.3 in Appendix F for the full computation).

Thus $P(\hat{\tau} \in \mathcal{B}) \leq A + B \leq A' + B'$. However, by our choice of η , we have that $A' \leq B'$. So,

$$P(\hat{\tau} \in \mathcal{B}) \leq 2N e^{-\eta^2 k/4\omega} \stackrel{(d)}{\leq} \frac{2}{N^3} \leq 2/\sqrt{k}, \quad (6)$$

where (d) follows from the bound $\eta \geq 4\sqrt{\omega} \sqrt{\frac{\log N}{k}}$. Now that we have bounded $P(\hat{\tau} \in \mathcal{B})$, we will bound $\mathbb{E}[w(\sigma(\hat{\tau}) \cap Z) \mid \hat{\tau} \notin \mathcal{B}]$. To that end, define $H = \{a_1, \dots, a_{(1-2\eta)k/\omega}\}$ to be the set of elements with the largest $(1 - 2\eta)k/\omega$ scores in $\mathcal{P}$, and let $\widehat{\text{OPT}} = w(H)$. We define a re-scoring of the elements of $\mathcal{P}$, where all elements except those in H have score 0: $\hat{w}(a) = \begin{cases} w(a) & \text{if } a \in H \\ 0 & \text{otherwise.} \end{cases}$ It follows that $\mathbb{E}[\hat{w}(\sigma(\hat{\tau})) \mid \hat{\tau} \notin \mathcal{B}] \geq \widehat{\text{OPT}}$, since H is a subset of $\sigma(\tau)$ for any $\tau \notin \mathcal{B}$. To prove the claim, it suffices to upper bound the total score of elements of H that appear in Y . In particular,

$$\mathbb{E}[\hat{w}(\sigma(\hat{\tau}) \cap Y) \mid \hat{\tau} \notin \mathcal{B}] \leq \mathbb{E}[w(H \cap Y)] = \frac{\widehat{\text{OPT}}}{2}.$$

$$\begin{aligned}
\text{In sum, we have that: } \mathbb{E}[w(\sigma(\hat{\tau}) \cap Z) \mid \hat{\tau} \notin \mathcal{B}] &\geq \mathbb{E}[\hat{w}(\sigma(\hat{\tau}) \cap Z) \mid \hat{\tau} \notin \mathcal{B}] \\
&= \mathbb{E}[\hat{w}(\sigma(\hat{\tau})) \mid \hat{\tau} \notin \mathcal{B}] - \mathbb{E}[\hat{w}(\sigma(\hat{\tau}) \cap Y) \mid \hat{\tau} \notin \mathcal{B}] \\
&\geq \widehat{\text{OPT}} - \frac{\widehat{\text{OPT}}}{2} = \frac{\widehat{\text{OPT}}}{2} \geq (1 - 2\eta) \frac{\text{OPT}}{2\omega}. \tag{7}
\end{aligned}$$

With this in mind, we can proceed to bounding the performance of the algorithm on Z .

$$\begin{aligned}
\mathbb{E}[w(\text{Alg}(Z))] &\geq \left(1 - \frac{2}{\sqrt{k}}\right) \mathbb{E}[w(\text{Alg}(Z)) \mid \hat{\tau} \notin \mathcal{B}] && \text{by (6)} \\
&\geq \left(1 - \frac{2}{\sqrt{k}}\right) \mathbb{E}[w(\sigma(\hat{\tau}) \cap Z) \mid \hat{\tau} \notin \mathcal{B}] \\
&\geq \left(1 - \frac{2}{\sqrt{k}}\right) \left(1 - 2D\sqrt{\frac{\log N}{k}}\right) \frac{\text{OPT}}{2\omega} && \text{by (7)} \\
&\geq \left(1 - \frac{2 + 2D\sqrt{\log N}}{\sqrt{k}}\right) \frac{\text{OPT}}{2\omega} = \left(\frac{1}{2} - \frac{1 + D\sqrt{\log N}}{\sqrt{k}}\right) \frac{\text{OPT}}{\omega}.
\end{aligned}$$

In sum,

$$\begin{aligned}
\mathbb{E}[w(\text{Alg})] &\geq \left[\frac{1}{2} - \frac{1 + 2\sqrt{2}C\sqrt{\log n}}{4\sqrt{k}} + \frac{1}{2} - \frac{1 + D\sqrt{\log N}}{\sqrt{k}}\right] \frac{\text{OPT}}{\omega} \\
&= \left[1 - \frac{5 + (4D + 2\sqrt{2}C)\sqrt{\log N}}{4\sqrt{k}}\right] \frac{\text{OPT}}{\omega}.
\end{aligned}$$

For the chosen $C = \frac{7+4D}{4-2\sqrt{2}}$, we have that $(4 - 2\sqrt{2})C - 4D = 7 \geq \frac{5}{\sqrt{\log 2}} \geq \frac{5}{\sqrt{\log N}}$. It follows that $[(4 - 2\sqrt{2})C - 4D]\sqrt{\log N} \geq 5$, which in turn gives that $5 + (4D + 2\sqrt{2}C)\sqrt{\log N} \leq 4C\sqrt{\log N}$. We can thus continue the above computation:

$$\mathbb{E}[w(\text{Alg})] \geq \left[1 - \frac{5 + (4D + 2\sqrt{2}C)\sqrt{\log N}}{4\sqrt{k}}\right] \frac{\text{OPT}}{\omega} \geq \left[1 - \frac{C\sqrt{\log N}}{\sqrt{k}}\right] \frac{\text{OPT}}{\omega}, \text{ as desired. } \quad \square$$

Note that Proposition 5.5 bounds the competitive ratio in the case where a bound on the width of the poset is known. Asymptotically, however, if k grows quickly enough, this assumption can be dropped, and a tight algorithm can be achieved (see Corollary 5.2).

COROLLARY 5.2 *Let $\mathcal{P}$ be a poset of width $\omega \leq \log k$. Algorithm 3 run with $\eta(N, k) = \sqrt{\frac{16 \log k \log N}{k}}$ (i.e., with width bound $\omega' = \log k$) is $\omega \left(1 - \frac{38 \log N}{\sqrt{k}}\right)^{-1}$ -competitive. Thus, if $(\mathcal{P}_N)_{N \geq 1}$ is a sequence of posets of width at most ω with $|\mathcal{P}_N| = N$ and $k = k(N)$ is such that $\log(N)/\sqrt{k} \rightarrow 0$, the competitive ratio of Algorithm 3 run with $\eta(N, k) = \sqrt{\frac{16 \log k \log N}{k}}$ is $\omega(1 + o(1))$ as $N \rightarrow \infty$.*

Proof. Using the notation from Proposition 5.5, we have that $D = 4 \log k$ and $C = \frac{7+4D}{4-2\sqrt{2}} \leq 38 \log k$. Since $k \rightarrow \infty$, we can restrict our attention to posets $\mathcal{P}_N$ for which $\log k(N) \geq \omega$. In this case, the reciprocal of the competitive ratio of the algorithm on $\mathcal{P}_N$ is at least

$$\frac{1}{\omega} \left(1 - \frac{38 \sqrt{\log k} \sqrt{\log N}}{\sqrt{k}}\right) \geq \frac{1}{\omega} \left(1 - \frac{38 \log N}{\sqrt{k}}\right).$$

Thus, if $\log(N)/\sqrt{k} \rightarrow 0$ as $N \rightarrow \infty$, then the competitive ratio is $\omega(1 + o(1))$. $\square$

6. Selection under the Group Model

Recall that under group bias, elements are partitioned into disjoint groups $G_1, \dots, G_g$, and the observed scores are $\tilde{w}(a) = w(a)/\beta_j$ if $a \in G_j$ (see Ex. 3.1). These group-specific bias factors imply comparability within each group and incomparability between groups. We can thus cast group bias as a special case of poset bias, wherein the poset is composed of disjoint chains, and the width is the number g of groups. Since g is known in advance, Corollary 5.1 implies that PROXYLABEL is $(g+1)e^2$ -competitive, which is an improvement over the $2e^3(2g+1)(1+o(1))$ bound obtained by POSETLABEL. We next provide a general framework for extending vanilla k -secretary algorithms to obtain better competitive ratios in the presence of group bias (in Section 6.1), using which we can, e.g., get a $g(1 - 5(k/g)^{-1/2})^{-1}$ -competitive algorithm by parallelizing the algorithm of Kleinberg (2005). After that, in Section 6.2, we demonstrate further improvements in the competitive ratio when scores and group assignments are stochastic.

6.1. A framework for extending vanilla algorithms to account for group bias

As mentioned above, our methods for the poset setting can be applied to the group setting as well. However, the structure of the poset in the group setting is quite simple (it is a disjoint union of chains), so we might expect simpler algorithms and better competitive ratios compared to the poset setting. Indeed, there is room for improvement, since PROXYLABEL is not tight compared to the lower bound in Proposition 5.1, and ADATHRESHOLD is only tight when $(\log N)/\sqrt{k} \rightarrow 0$.

One natural idea is to apply a vanilla k -secretary algorithm on each group, selecting at most k/g elements from each. Since secretary algorithms take the number of elements N as input, a pure parallelization over groups would require advance knowledge of the group sizes. However, one can avoid this issue by only parallelizing phases of the algorithm that do not make use of N . To make this explicit, we define a class of *independent-sample secretary (ISS) algorithms*, which do not use knowledge of N during their selection phases.

Definition 6.1 (independent-sample secretary (ISS) algorithm) *A k -secretary algorithm $\mathcal{A}^c(N, k)$ which takes as input the number of elements N and the selection capacity k , and operates on a stream $\mathcal{C}$ of elements, is an independent-sample secretary (ISS) algorithm if the following hold for some $p_k \in [0, 1]$:*

1. *$\mathcal{A}$ can be temporally decomposed into a **sampling phase** $\mathcal{A}_{\text{sample}}^c(N, k)$ which returns a sample $S \subseteq \mathcal{C}$, followed by a **selection phase** $\mathcal{A}_{\text{select}}^{c \setminus S}(S, k)$ on the remaining elements $\mathcal{C} \setminus S$; and*
2. *the sampling phase returns the first $\text{Bin}(N, p_k)$ elements and makes no selections.*

Note that the sampling phase of an ISS algorithm uses the number N of elements, which prevents us from parallelizing this phase in a straightforward way. To get around this, we run the selection

Algorithm 4: GROUP-AWARE PARALLELIZATION (GAP) for (adversarial) group bias.

input: ISS algorithm $\mathcal{A}(N, k)$, N , k , number of groups g , stream of elements $\mathcal{C}$

- 1 Produce a sample S using the sampling phase $\mathcal{A}_{\text{sample}}^c(N, k/g)$
 - 2 Run the selection phase $\mathcal{A}_{\text{select}}^{c \cap G_j \setminus S}(S \cap G_j, k/g)$ for each $j \in [g]$
-

phase on the entire stream (unparallelized) and form group-specific samples by intersecting the unparallelized sample with each group. On the other hand, since ISS algorithms do not make use of N during their selection phases, we are free to directly parallelize these selection phases. For example, the k -secretary algorithm of Kleinberg (2005), the matroid¹⁷ secretary algorithm of Feldman et al. (2015), and the graphic matroid secretary algorithm of Soto et al. (2018) can all be viewed as ISS algorithms. The resulting algorithm (Algorithm 4) preserves the competitive ratio of the vanilla algorithm up to a factor of g .

PROPOSITION 6.1 *Let $\mathcal{A}$ be an $f(k)$ -competitive ISS algorithm for the vanilla k -secretary problem satisfying ordinal fairness¹⁸ with respect to the total order on elements. Then GAP (Alg. 4) using $\mathcal{A}$ is $g \cdot f(k/g)$ -competitive for the k -secretary problem under group bias. Moreover, GAP satisfies ordinal fairness with respect to the group poset.*

Proof. The crux of this proof is that the GAP is *distributionally equivalent* to running the vanilla algorithm on each group separately, with a selection capacity of k/g for each group. More formally, we claim that GAP induces the same sampling and selection probabilities as running $\mathcal{A}^{c \cap G_j}(|G_j|, k/g)$ independently on each group G_j .

To verify this, it suffices to show that sampling probabilities are preserved, since the selection phases are run independently of each other. To that end, note that $\mathbb{P}(a \in S) = p_{k/g}$ regardless of the group membership of a . Thus, the sampling distribution induced by GAP on G_j is the same as that of $\mathcal{A}^{c \cap G_j}(|G_j|, k/g)$. We can thus analyze each group separately. Letting $\text{Alg}^{c \cap G_j}(|G_j|, k/g)$ denote the output of $\mathcal{A}^{c \cap G_j}(|G_j|, k/g)$, we have

$$\mathbb{E}[w(\text{Alg}_4)] = \sum_{j=1}^g \mathbb{E}\left[w(\text{Alg}^{c \cap G_j}(|G_j|, k/g))\right] \geq \sum_{j=1}^g \frac{\text{OPT}_{k/g}(G_j)}{f(k/g)} \geq \frac{\text{OPT}_{k/g}}{f(k/g)} \geq \frac{\text{OPT}_k}{gf(k/g)},$$

where $\text{OPT}_m(G)$ is the sum of the top m scores in group G , and OPT_m is the sum of the top m scores overall. The parallelized algorithm therefore has competitive ratio at most $gf(k/g)$. $\square$

¹⁷ The matroid secretary setting can again be extended to include bias models, i.e., every element is assumed to have an underlying group that they belong to (e.g., one can call it a poset matroid secretary problem). The goal is then to maximize the true total utility of selected elements, constrained to be an independent set in the matroid.

¹⁸ In this case, ordinal fairness reduces to (i) increasing probability of selection with respect to increasing ranking, and (ii) decisions be made only with respect to the rank of the element rather than observed utility.

Algorithm 5: GAPCAP for the stochastic group setting

input: N , k , distribution over groups $\mathbf{p}$, quotas vector s

- 1 $S \leftarrow$ first $\lfloor \frac{N}{e} \rfloor$ elements; $R_j \leftarrow$ top $s_j k$ scores in $S \cap G_j$, for each $j \in [g]$
- 2 **for** $a \in G_j, a \notin S$ **do**
- 3 **if** $\tilde{w}(a) \geq \min R_j$ **then**
- 4 **if** the current $s_j k$ th best element from G_j was sampled **then** Select a .
- 5 Update R_j to be the $s_j k$ highest scores seen so far from G_j

Proposition 6.1 shows that a large class of secretary algorithms can be extended to the group bias setting while suffering only a factor g in competitive ratio. In particular, any tight ISS algorithm (i.e., one whose competitive ratio is $1 + o(1)$ as $k \rightarrow \infty$) such as the $(1 - 5k^{-1/2})^{-1}$ -competitive algorithm of Kleinberg (2005) can be extended to a tight algorithm (one whose competitive ratio is $g(1 + o(1))$ as $k \rightarrow \infty$) for the group bias setting.

6.2. Stochastic group assignment

We end our discussion on group bias by considering a stochastic setting, where each element is assigned independently to a group according to a probability distribution $\mathbf{p} = (p_1, \dots, p_g)$. Element scores are then sampled independently from the same nonnegative distribution. We assume that group assignments and score assignments are independent together as well.

Since the scores are stochastic, the top k elements will be proportionally distributed between the groups, on average. A natural idea would then be to apply the GAP framework, replacing the uniform quotas with proportional quotas of $p_j k$ for group G_j . To ensure that the selection probabilities are the same across groups,¹⁹ we will require the probability of an element in the sample to be independent of k (otherwise, the differing selection capacities would result in different selection probabilities). We give a simple example of how the k -secretary algorithm of Babaioff et al. (2007) can be parallelized, which we call GAPCAP. Due to the stochastic assumptions, the competitive ratio is constant, independent of g .

PROPOSITION 6.2 *Under the group bias setting, where groups are assigned according to the distribution $\mathbf{p}$ and scores are sampled independently from the same nonnegative distribution, GAPCAP on input of a proportions vector $s = \mathbf{p}$ is $2e(1 + o(1))$ -competitive; additionally, if $p_j < 1 - \frac{1}{k}$ for all $j \in [g]$, then the competitive ratio is bounded by $4e$. Moreover, GAPCAP satisfies OF.*

The factor of e in the competitive ratio comes from bounding the probability that one of the top-scoring elements in a group is selected by the algorithm. Since groups and scores are stochastic,

¹⁹ This quality is desired because it allows for a *single* sample to be taken, which is then intersected with each group to obtain group-specific samples.

the expected number of the top k elements in group G_j is $p_j k$, which means that our hedging is good on average. This is ultimately why we are able to avoid the factor of g in competitive ratio, as explained in more detail in the full proof in Appendix C. As with GAP, one drawback of this algorithm is that it imposes quotas, which limits its applicability.

7. Experiments

In this section, we compare the impact of (1) the vanilla adaptive threshold-based k -secretary algorithm of Kleinberg (2005), (2) the parallelized version of the same k -secretary algorithm using GAP, (3) POSETLABEL, and (4) ADATHRESHOLD on a real-world dataset. We develop a prediction model based on training data and use the resulting gender-specific errors to construct a partial ranking of the applicants. Using the model-predicted scores as a basis for the first two algorithms and the partial ranking as a basis for the second two, we compare selection rates across gender.

We use the Aspiring Minds' Employability Outcomes 2015 (AMEO 2015) dataset, which was created by Aspiring Minds, an organization which offers pre-employment assessments. We preprocess the data to only consider job seekers in computer science fields, and to only consider the following attributes: gender, standardized high school test scores, college tier, college GPA, college city tier, age at graduation, and Aspiring Minds test scores. 2212 data points remain after preprocessing, among which a random sample of 1600 is taken to be the training data. The other 612 data points comprise the test data, which includes 166 female and 446 male job seekers.²⁰

We take the Aspiring Minds computer programming test score as a proxy for utility ("true scores"), and as such, develop a model for predicting computer programming scores from the other attributes. In particular, we predict computer programming using linear regression.²¹ The coefficients of the linear regression model are given in Table 2 in Appendix G.²²

In the Aspiring Minds dataset, we see that the female and male score distributions are qualitatively similar ($\mu_{\text{female}} = 459.27, \sigma_{\text{female}} = 80.54, \mu_{\text{male}} = 477.99, \sigma_{\text{male}} = 93.613$) and are depicted in Figure 3 (top left). In order to compare different algorithms under different distributional settings, we transform the Aspiring minds datasets in two ways: decrease all female scores by 28.09 points (chosen so that the means differ by $\sigma_{\text{male}}/2$) to create a left-shifted distribution, and increase all

²⁰ We remark that gender is not binary; we make this dichotomy here because all candidates in the dataset were labeled as male or female.

²¹ We remark that any other machine learning model could have been used to predict performance as well, or the scores could have been obtained using an interview committee. This is simply an example of a predictive model that could be used in a hiring setting.

²² The linear regression model has a coefficient of -16.95 on the "gender" attribute, meaning that changing a candidate's gender from male to female would decrease the prediction by 16.95 points. One might wonder if a gender-blind model might be fairer in some sense. However, we find that omitting gender does not produce a significant difference in accuracy across groups.

female scores by 65.53 points (again, chosen so that the means differ by $\sigma_{\text{male}}/2$) to create a right-shifted distribution. See Figure 3 for a depiction of the true and predicted computer programming scores in the data, after training on unshifted (left-most) and shifted (center, right) data.

We generate score intervals for applicants (and thus partial rankings) by statistically comparing true scores and predicted scores by gender in the training data. In particular, suppose the true scores of Group j in the training data are in the vector y , and the predicted scores are $\hat{y}$. Then observe that $\hat{y}_{\text{transf.}} := \frac{\sigma_y}{\sigma_{\hat{y}}}(\hat{y} - \mu_{\hat{y}}) + \mu_y$ has the same mean and standard deviation as y . Now let σ_j denote the standard deviation of $y - \hat{y}_{\text{transf.}}$, which is a measure of error in these transformed scores. We can form score intervals in the test data as follows: given a predicted score $\tilde{w}(a)$ in Group j , we construct the interval

$$\left[\frac{\sigma_y}{\sigma_{\hat{y}}}(\tilde{w}(a) - \mu_{\hat{y}}) + \mu_y - \lambda\sigma_j, \frac{\sigma_y}{\sigma_{\hat{y}}}(\tilde{w}(a) - \mu_{\hat{y}}) + \mu_y + \lambda\sigma_j \right] = [\hat{y}_{\text{transf.}}(a) - \lambda\sigma_j, \hat{y}_{\text{transf.}}(a) + \lambda\sigma_j], \quad (8)$$

where $\lambda > 0$ is a parameter controlling the confidence in the score range. Note that under this construction, interval lengths will be uniform within any group, but may differ between groups. We call this the *error-correction approach*.

A total of twelve experiments were performed: the four algorithms were each run on the three distributions discussed above, each of which comprised of the 612 applicants in the test data. The vanilla algorithm adaptively increases a single threshold and makes selections above the current threshold, and GAP (Algorithm 4) parallelizes this algorithm. Each of these experiments was given a selection capacity of $k = 25$ and was run 10,000 times (see Appendix G for the same experiments run with $k = 50$). Since POSETLABEL has an extensive sampling phase compared to the other three algorithms, we scaled its sample size by $1/3$; in this way, the selection rates of POSETLABEL will be large enough to make meaningful comparisons to those of the other algorithms. Finally, we set the λ parameter in (8) to be $1/6$, which struck a nice balance between accounting for uncertainty and retaining enough ordinal information (experiments run with $\lambda = 1/3$ and $\lambda = 1/12$ produced qualitatively similar results, as discussed in Appendix G).

The performance of the vanilla k -secretary algorithm, POSETLABEL, and ADATHRESHOLD is sensitive to the true score distributions in each of the three scenarios: when the female true score distribution is low, fewer female applicants are selected, and when the female true score distribution is high, more female applicants are selected; GAP (parallelizing the algorithm of Kleinberg (2005)), on the other hand, uses quotas and is not sensitive to these distributional changes. Use of quotas might have been desirable if demographic parity was the goal, but its lack of sensitivity to scores can be undesirable given the legal backdrop of hiring in the U.S. and given its coarse, non-individualistic nature. PosetLabel and ADATHRESHOLD instead produce selection rates that are closer to each

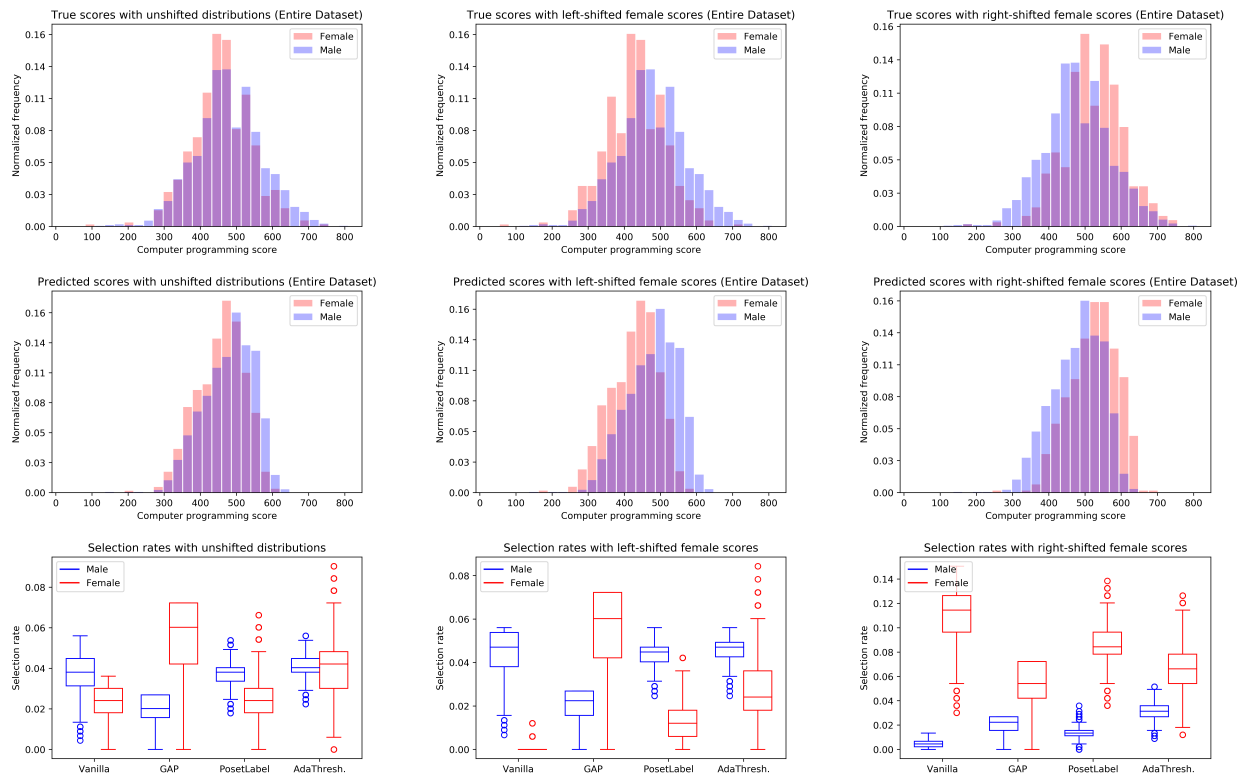


Figure 3 Experiment results for three “true score” distributions are shown: the unaltered programming scores (left), the distribution with left-shifted female scores (center column), and the distribution with right-shifted female scores (right). The true score distributions are shown by gender (top), as well as the predicted scores from the regression model (center row) and the selection rates of three algorithms (bottom).

other (thereby giving some benefit of the doubt to the group with higher uncertainty in their evaluation) compared to the vanilla algorithm as well as GAP. Since both these algorithms use posets derived from errors in prediction, they are better justified from a legal standpoint as well.

Biases enhanced by prediction models. The above experiments showed a natural adaptiveness in selection rates as the underlying data was altered in the three scenarios considered. This adaptiveness can be attributed to the trends in the underlying data captured by the prediction models. However, if the prediction model carries a definitive bias (e.g., a trend not seen in underlying data but picked up by the ML model) against some group, then algorithms making decisions based on the predictive model will be impacted. We next show that, in fact, when score intervals are constructed as per the error-correction approach (equation (8)), such group-specific biases can be corrected using our proposed poset-based algorithms.

To demonstrate this, we artificially bias the prediction model as follows: we first center the underlying data so that both groups have the same mean in the training data, and then fit a linear regression model to predict CS scores. We artificially decrease the gender coefficient in the best

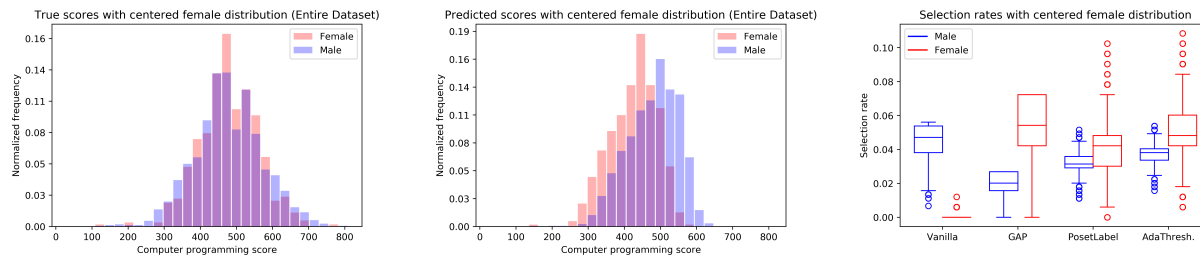


Figure 4 The four algorithms are run on centered data (i.e., the true scores of the female group were translated so that both groups have the same mean true score). The regression model was then modified to introduce an additive bias of 50 points against the female group. Shown above are the true score distributions (left), the predicted score distributions (center), and the resulting selection rates (right).

fit linear regression model by 50 points, thus introducing a (definitive) additive bias against the female group. We ran the four algorithms on the test data using the shifted predictions.

We find that since the vanilla algorithm has no way of accounting for this additional ML-model bias, it results in a very low selection rate for female applicants. The POSETLABEL and ADATHRESHOLD algorithms, on the other hand, are able to account for this additional ML-model bias by the error-correction approach (eq. (8)) and therefore incorporate the knowledge of the true scores in the training data. The selection rates produced by these algorithms are consequently more appropriate (given the true score distributions) than that of the vanilla algorithm, as shown in Figure 4. This experiment reiterates that the poset-based approaches can be sensitive to changing trends in data and can mitigate the impact of various sources of bias.

8. Conclusion

Selecting a deserving set of qualified people is an incredibly important yet complex problem in practice. This work is the first to formally consider bias in data using partially ordered sets, and the first to consider bias within the context of the secretary problem. For the general case of partial ordinal rankings, we provided two algorithms: one with an order-optimal competitive ratio (POSETLABEL), and another using adaptive thresholds with a tight competitive ratio (ADATHRESHOLD). These algorithms make selections based solely on the partial rankings and arrival order,²³ which means that properties of the poset (say, those designed to mitigate bias) directly affect selection decisions. We additionally provided simpler algorithms for the special case of group bias, where the poset has simpler structure and vanilla secretary algorithms can be parallelized across the groups. Our key takeaway is the following: *accounting for uncertainty and bias in algorithmic decision-making can improve the quality of selected applicants and produce fairer decisions.*

The necessity for such bias-aware methods is born in part out of the disparate amount of errors in evaluation metrics on applicants, and our analysis shows that poset-based algorithms constitute

²³ Thus, by Proposition 3.1, they satisfy ordinal fairness.

interventions that provably increase the true total utility of applicants selected. Many of the ideas presented here can be adapted to other data-driven models where there is reason to believe that underlying data is biased. Our algorithmic approaches are also versatile: (1) the random partitioning technique of POSETLABEL can be adapted to a non-algorithmic setting by forming independent review committees, each of which reviews a random subset of the applicant pool. Such a process can reduce the impact of a single reviewer who may dominate a debrief; (2) enforcing diversity in a selected pool of candidates (GAP) might be desired for certain applications such as appointment committees in academic review boards; and (3) the technique of adaptive thresholding used in ADATHRESHOLD can be applied generally in practice. For example, in a rolling decisions scenario, where there is no bound on the number of applicants over time, one can periodically update the threshold based on the available data; additionally, if distributions of scores are presumed to shift over time, thresholds can be computed using a sliding window to ensure up-to-date selection criteria. These managerial considerations can positively impact hiring practices.

Secretary problems operate under many assumptions, such as access to a random sample of applicants and the knowledge of the total number of applicants. In some cases, these assumptions can be justified in practice; e.g., data on past applicants could act as the random sample, and time constraints in hiring can upper bound the number of applicants that can be reviewed. Theoretically speaking though, designing algorithms which avoid rejecting elements “in sample sets” is an open question; an LP-based workaround for this issue exists for the unbiased $k = 1$ case, but remains open for larger k (Buchbinder et al. 2014). Determining an optimal competitive ratio without the knowledge of N also remains an open question in both the unbiased and biased settings, although Gharan and Vondrák (2011) showed an N' -dependent lower bound²⁴ on competitive ratio for the classical 1-secretary problem when N is chosen adversarially from the set $\{1, \dots, N'\}$.

Some of our algorithms (GAPCAP and GAP) make use of group-specific selection capacities (i.e., *quotas*). Legally speaking, demographic quotas may or may not be admissible in certain situations (e.g., (Thomas 2016, Duffy 2020)). Whether such a disparate treatment can be justified in practice by the inadequacies in data/evaluations is an open question for the legal scholars. On the other hand, if the poset contains only justifiable comparative information amongst candidates and gives no leeway simply due to group membership, one does not have to adhere to quotas to make “fair” selections. We believe such an approach is attractive from a practical, ethical, and legal perspectives as well (e.g., see Salem et al. (2022)). Further, since our competitive ratios depend on the width of the poset, firms have an incentive to invest in more information about the candidates’ true potentials (e.g., by conducting more interviews or summer training programs) which can ultimately help reduce the width of the poset.

²⁴ In particular, the lower bound they found was the N' -th harmonic number.

References

- AspiringMinds. 22.05.2020. Aspiring Minds: AI powered talent evaluation. <https://www.aspiringminds.com/>
- Moshe Babaioff, Michael Dinitz, Anupam Gupta, Nicole Immorlica, and Kunal Talwar. 2009. Secretary problems: weights and discounts. In *Proceedings of the twentieth annual ACM-SIAM Symposium on Discrete Algorithms*. SIAM, 1245–1254.
- Moshe Babaioff, Nicole Immorlica, David Kempe, and Robert Kleinberg. 2007. A knapsack secretary problem with applications. In *Approximation, randomization, and combinatorial optimization. Algorithms and techniques*. Springer, 16–28.
- Allan Borodin and Ran El-Yaniv. 1998. *Online Computation and Competitive Analysis*. Cambridge Univ. Press, USA.
- Niv Buchbinder, Kamal Jain, and Mohit Singh. 2014. Secretary problems via linear programming. *Mathematics of Operations Research* 39, 1 (2014), 190–206.
- Marilyn Cavicchia. 2015. How to fight implicit bias? With conscious thought, diversity expert tells NABE. American Bar Association: Bar Leader.
- L Elisa Celis, Anay Mehrotra, and Nisheeth K Vishnoi. 2020. Interventions for ranking in the presence of implicit bias. In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*. 369–380.
- Stephen Coate and Glenn Loury. 1993. Will Affirmative-Action Policies Eliminate Negative Stereotypes? *American Economic Review* 83 (02 1993), 1220–40.
- Sam Corbett-Davies, Emma Pierson, Avi Feller, Sharad Goel, and Aziz Huq. 2017. Algorithmic decision making and the cost of fairness. In *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. ACM, 797–806.
- Maria De-Arteaga, Alexey Romanov, Hanna Wallach, Jennifer Chayes, Christian Borgs, Alexandra Chouldechova, Sahin Geyik, Krishnaram Kenthapadi, and Adam Tauman Kalai. 2019. Bias in bios: A case study of semantic representation bias in a high-stakes setting. In *Proceedings of the Conference on Fairness, Accountability, and Transparency*. 120–128.
- Claire Duffy. 2020. Plans at Microsoft and Wells Fargo to increase Black leadership are under scrutiny from the Labor Dept. *CNN Business* (October 7, 2020). <https://www.cnn.com/2020/10/07/business/microsoft-wells-fargo-diverse-hiring-probe/index.html>
- Cynthia Dwork, Moritz Hardt, Toniann Pitassi, Omer Reingold, and Richard S. Zemel. 2011. Fairness through awareness. In *ITCS '12*.
- E. B. Dynkin. 1963. The optimum choice of the instant for stopping markov process. *Soviet Mathematics, Doklady* 4.

- Vitalii Emelianov, Nicolas Gast, Krishna P Gummadi, and Patrick Loiseau. 2020. On Fair Selection in the Presence of Implicit Variance. In *Proceedings of the 2020 ACM Conference on Economics and Computation (EC 2020)*.
- Moran Feldman, Ola Svensson, and Rico Zenklus. 2015. A Simple $O(\log \log(\text{rank}))$ -competitive Algorithm for the Matroid Secretary Problem. In *Proc. of the Twenty-sixth Annual ACM-SIAM Symp. on Discrete Algorithms (SODA '15)*. Society for Industrial and Applied Mathematics, Philadelphia, PA, USA, 1189–1201.
- Shayan Oveis Gharan and Jan Vondrák. 2011. On Variants of the Matroid Secretary Problem. *CoRR* abs/1104.4081 (2011).
- Rachel Goodman. 2018. Why Amazon’s Automated Hiring Tool Discriminated Against Women. <https://www.aclu.org/blog/womens-rights/womens-rights-workplace/why-amazons-automated-hiring-tool-discriminated-against> Published Oct. 12, 2018. Last accessed Jun. 6, 2019.
- Spencer Greenberg and Mehryar Mohri. 2014. Tight lower bound on the probability of a binomial exceeding its expectation. *Statistics & Probability Letters* 86 (2014), 91–98.
- Matthew Joseph, Michael Kearns, Jamie H Morgenstern, and Aaron Roth. 2016. Fairness in learning: Classic and contextual bandits. *Advances in neural information processing systems* 29 (2016).
- Henry Kierstead and William Trotter. 1981. An extremal problem in recursive combinatorics. *Proceedings of the Twelfth Southeastern Conference on Combinatorics, Graph Theory and Computing* Vol. II (1981).
- Jon M. Kleinberg and Manish Raghavan. 2018. Selection Problems in the Presence of Implicit Bias. In *9th Innovations in Theoretical Computer Science Conference, ITCS 2018, January 11-14, 2018, Cambridge, MA, USA*. 33:1–33:17. <https://doi.org/10.4230/LIPIcs.ITCS.2018.33>
- Robert Kleinberg. 2005. A Multiple-choice Secretary Algorithm with Applications to Online Auctions. In *Proceedings of the Sixteenth Annual ACM-SIAM Symposium on Discrete Algorithms (SODA '05)*. Society for Industrial and Applied Mathematics, Philadelphia, PA, USA, 630–631. <http://dl.acm.org/citation.cfm?id=1070432.1070519>
- Ravi Kumar, Silvio Lattanzi, Sergei Vassilvitskii, and Andrea Vattani. 2011. Hiring a Secretary from a Poset. *Proceedings of the ACM Conference on Electronic Commerce*, 39–48.
- Jeff Larson, Surya Mattu, Lauren Kirchner, and Julia Angwin. 2016. How We Analyzed the COMPAS Recidivism Algorithm. *ProPublica* (2016).
- Michael J Lovaglia, Jeffrey W Lucas, and Shane R Thye. 1998. Status processes and mental ability test scores. *Amer. J. Sociology* 104, 1 (1998), AJSv104p195–228.
- Kristian Lum. 2016. Predictive Policing Reinforces Police Bias. *Human Rights Data Analysis Group* (2016).
- Corinne Moss-Racusin, Dovidio, Victoria Brescoll, Mark Graham, and Jo Handelsman. 2012. Science Faculty’s Subtle Gender Biases Favor Male Students. *Proceedings of the National Academy of Sciences* 109 (09 2012), 16474–16479.

- Hussein Mouzannar, Mesrob I. Ohannessian, and Nathan Srebro. 2019. From Fair Decision Making To Social Equality. In *Proceedings of the Conference on Fairness, Accountability, and Transparency (FAT* '19)*. ACM, New York, NY, USA, 359–368.
- Emma Pierson, Sam Corbett-Davies, Camelia Simoiu, Vignesh Ramachandran, Sharad Goel, Jan Overgoor, and Cheryl Phillips. 2019. A large-scale analysis of racial disparities in police stops across the United States. *The Stanford Open Policing Project* (2019).
- Manish Raghavan, Solon Barocas, Jon Kleinberg, and Karen Levy. 2020. Mitigating bias in algorithmic hiring: Evaluating claims and practices. In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*. 469–481.
- Manish Raghavan, Solon Barocas, Jon M. Kleinberg, and Karen Levy. 2019. Mitigating Bias in Algorithmic Employment Screening: Evaluating Claims and Practices. *CoRR* abs/1906.09208 (2019). arXiv:1906.09208 <http://arxiv.org/abs/1906.09208>
- Jad Salem, Deven R Desai, and Swati Gupta. 2022. Don't let Ricci v. DeStefano Hold You Back: A Bias-Aware Legal Solution to the Hiring Paradox. *arXiv preprint arXiv:2201.13367* (2022).
- Ray Sanchez. 2022. The NFL made a rule to boost diversity in upper ranks. It's not what 'my father intended it to be'. <https://www.cnn.com/2022/02/03/sport/nfl-brian-flores-lawsuit-rooney-rule/index.html> Published Feb. 4, 2022. Last accessed Mar. 17, 2022.
- Javier Sánchez-Monedero, Lina Dencik, and Lilian Edwards. 2020. What Does It Mean to “solve” the Problem of Discrimination in Hiring? Social, Technical and Legal Perspectives from the UK on Automated Hiring Systems. In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency (FAT* '20)*. Association for Computing Machinery, New York, NY, USA, 458–468. <https://doi.org/10.1145/3351095.3372849>
- J. Soto. 2013. Matroid Secretary Problem in the Random-Assignment Model. *SIAM J. Comput.* 42, 1 (2013), 178–211.
- José A Soto, Abner Turkieltaub, and Victor Verdugo. 2018. Strong algorithms for the ordinal matroid secretary problem. In *Proceedings of the Twenty-Ninth Annual ACM-SIAM Symposium on Discrete Algorithms*. SIAM, 715–734.
- Steven Spencer, Claude M. Steele, and Diane M. Quinn. 1999. Stereotype Threat and Women's Math Performance. *Journal of Experimental Social Psychology* 35 (1 1999), 4–28. Issue 1.
- Claude M. Steele and J. Aronson. 1995. Stereotype threat and the intellectual test performance of African Americans. *Journal of Personality and Social Psychology* 69 (1995), 797–811. Issue 5.
- Tracy Thomas. 2016. Reconsidering the Remedy of Gender Quotas. *Harvard Journal of Law & Gender Online* (2016).

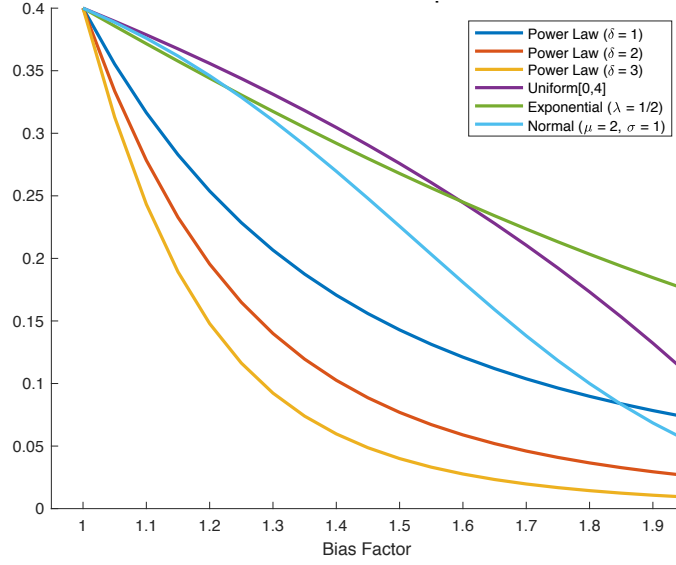


Figure 5 Ratio of elements selected from group G_2 , assuming group bias with two groups, over varying β_2 . Group membership is sampled from $(.6, .4)$, $\beta_1 = 1$, true scores are stochastic, and $k = \infty$ (see Prop. A.1).

Appendix A: Missing proofs of Section 4

PROPOSITION A.1 Suppose the score $w(a)$ of each element is drawn independently from some distribution $\mathcal{D}$. Suppose for each element a , we have $a \in G_1$ with probability $p \in (0, 1)$ and $a \in G_2$ otherwise. The observed scores from G_2 experience a bias $\beta_2 = \beta > 1$, and G_1 elements experience no bias. Consider a policy that selects elements if their observed score is at least τ (for some arbitrary threshold τ), with no limit on the number of selections. Then the ratio of the expected number of selected G_2 elements to the expected number of selected elements is $\frac{(1-p)\mathbb{P}(w(a) \geq \beta\tau)}{(1-p)\mathbb{P}(w(a) \geq \beta\tau) + p\mathbb{P}(w(a) \geq \tau)}$. In particular, this proportion tends to 0 as $\beta \rightarrow \infty$.

Proof of Proposition A.1. Let F denote the cumulative distribution function of $\mathcal{D}$. Then an element from G_1 is selected with probability $1 - F(\tau)$, and an element from G_2 is selected with probability $1 - F(\beta\tau)$, as

$$F(\beta\tau) = \mathbb{P}(\{w(a) \leq \beta\tau\}) = \mathbb{P}(\{\tilde{w}(a) \leq \tau\}) .$$

Letting S_{G_1} (resp. S_{G_2}) denote the set of selected elements from G_1 (resp. G_2), we have that

$$\mathbb{E}(|S_{G_2}|) = N(1-p)(1 - F(\beta\tau))$$

and similarly, $\mathbb{E}(|S_{G_1}|) = Np(1 - F(\tau))$. So, the ratio of $\mathbb{E}(|S_{G_2}|)$ to $\mathbb{E}(|S_{G_1} \cup S_{G_2}|)$ is

$$\frac{(1-p)(1 - F(\beta\tau))}{(1-p)(1 - F(\beta\tau)) + p(1 - F(\tau))} .$$

Note that as $\beta \rightarrow \infty$, we have that $(1-p)(1 - F(\beta\tau)) \rightarrow 0$. So, the above ratio tends to 0 as $\beta \rightarrow \infty$, assuming $F(\tau) \neq 1$. $\square$

Algorithm 6: K-SECRETARY (Babaioff et al. 2007)

input: N, k

- 1 $S \leftarrow$ first $\lfloor \frac{N}{e} \rfloor$ elements
- 2 $R \leftarrow$ top k scores seen so far
- 3 **for** $a \notin S$ **do**
- 4 **if** $w(a) \geq \min R$ **then**
- 5 **if** the current k th highest score corresponds to some element in S , **then** select a
- 6 Update R to be the k highest scores seen so far

Next, we discuss the performance of bias-agnostic algorithms in the case of group bias. Under group bias, any bias-agnostic secretary algorithm has an intuitive shortcoming: selection decisions are made based on inaccurate data. Therefore, any algorithm that selects elements if their observed score is higher than a fixed threshold, will underselect elements from a group that faces bias (see Proposition A.1 above for a discussion of this phenomenon for randomly generated scores). The next proposition quantifies the extent to which the presence of group bias can increase an algorithm's competitive ratio.

PROPOSITION A.2 K-SECRETARY (Algorithm 6) does not satisfy ordinal fairness under group bias.

Proof. Consider an instance with $k = 2$ and two groups of equal size, $|G_1| = |G_2| = N/2$. Suppose that $\beta_1 = 1$, and β_2 is large enough that

$$\tilde{w}(a) > \tilde{w}(b) \quad \text{whenever } a \in G_1, b \in G_2 .$$

Now let a be the lowest-ranked element in G_1 , and b the lowest-ranked element in G_2 . Then

$$0 = \mathbb{P}(\{b \text{ is selected}\}) < \mathbb{P}(\{a \text{ is selected}\}) ,$$

and so ordinal fairness is violated. $\square$

PROPOSITION A.3 For each element a , let $[\beta_a^\ell, \beta_a^u] \subset [1, \infty)$ be the range of bias factors given to the algorithm, and suppose $\mathcal{A}$ is α -competitive for the vanilla k -secretary problem. Now suppose that for each a , we choose a score $\hat{w}(a)$ which lies in the appropriate interval $[\beta_a^\ell \tilde{w}(a), \beta_a^u \tilde{w}(a)]$. Then running $\mathcal{A}$ on the chosen scores $\hat{w}$ is an (αM^2) -competitive algorithm, where $M = \max_a \frac{\beta_a^u}{\beta_a^\ell}$.

Proof of Proposition A.3. Let $w(a_1) \geq \dots \geq w(a_k)$ be the k highest scores, and let $\hat{w}(b_1) \geq \dots \geq \hat{w}(b_k)$ be the k highest chosen scores. Note that for any a , we have

$$\frac{w(a)}{M} \leq \underbrace{\tilde{w}(a)\beta_a^\ell}_{= \ell_a} \leq w(a), \hat{w}(a) \leq \underbrace{\tilde{w}(a)\beta_a^u}_{= u_a} \leq Mw(a) ,$$

where $M = \max_a \beta_a^u / \beta_a^\ell$. Letting Alg denote the output of $\mathcal{A}$ run on $\hat{w}$, this implies

$$\mathbb{E}(w(\text{Alg})) \geq \frac{1}{M} \mathbb{E}(\hat{w}(\text{Alg})) \geq \frac{1}{\alpha M} \sum_{i=1}^k \hat{w}(b_i) \geq \frac{1}{\alpha M} \sum_{i=1}^k \hat{w}(a_i) \geq \frac{1}{\alpha M^2} \sum_{i=1}^k w(a_i),$$

as desired. $\square$

Next, we show that the K-SECRETARY is $\Omega(M)$ -competitive when it chooses arbitrary weights (within the appropriate intervals). Consider the following instance with N elements and a selection capacity of $k = 1$:

$$w(a_i) = \begin{cases} \beta & i = 1 \\ 1 + \varepsilon & i = 2, \dots, N \end{cases}, \quad \text{and } \beta_{a_1} = \beta, \beta_{a_i} = 1 \text{ for } i = 2, \dots, N.$$

So, the observed scores by the algorithm are

$$\tilde{w}(a_i) = \begin{cases} 1 & i = 1, \\ 1 + \varepsilon & i = 2, \dots, N. \end{cases}$$

Suppose the interval of bias factors given for a_1 is $[1, \beta]$, and the intervals given for all other elements are $[1, 1]$. Then the algorithm could choose the minimum bias factor for each element, in which case the chosen scores are

$$\hat{w}(a_i) = \begin{cases} 1 & i = 1, \\ 1 + \varepsilon & i = 2, \dots, N. \end{cases}$$

In this case, the algorithm will certainly choose an element of score $1 + \varepsilon$, and so

$$\frac{\mathbb{E}(w(\text{Alg}))}{\text{OPT}} = \frac{1 + \varepsilon}{\beta}.$$

Letting $\varepsilon \rightarrow 0$, we see that K-SECRETARY is $\Omega(M)$ -competitive.

Appendix B: Ordinal fairness and conditional demographic parity

Here, we formalize the relationship between ordinal fairness and conditional demographic parity alluded to in Section 3.2. Suppose there is a predictive attribute X which takes values in $\{1, 2, 3, 4, 5\}$, and you would like to make fair decisions with respect to X across demographic groups. Conditional demographic parity would require that $\mathbb{P}(Y = 1 \mid X = x, G = g) = \mathbb{P}(Y = 1 \mid X = x)$ for any g , where Y is the binary decision and G is the group membership.

One can construct a partial order on the elements by $a_i \prec a_j$ if and only their X values satisfy $X(a_i) < X(a_j)$. See Figure 6 for an example of this construction on eight elements. Note that under this construction, any two elements with the same X value are order isomorphic. Thus under ordinal fairness, any two elements in the same horizontal band in Figure 6 must be treated equally. Ordinal fairness thus directly implies conditional demographic parity for this construction, as elements with the same X value must be treated equally.

Appendix C: Missing analysis from Section 6

Proof of Proposition 6.2 Recall that each element is assigned to a group according to the probability distributions $\mathbf{p} = (p_1, \dots, p_g)$, and the algorithm applies proportional quotas $sk = \mathbf{p}k$. Let OPT_p denote the optimal parallelized solution; that is, the set of the top $p_j k$ elements from group G_j , for $j \in [g]$.

We begin by showing that each element in OPT_p is selected with probability at least e^{-1} . To see this, suppose $a \in \text{OPT}_p$, and without loss of generality, $a \in G_1$. Then a is selected if and only if it arrives after the sample, and the $p_1 k$ th best element from G_1 seen so far was in the sample. Note that the probability of any particular element being sampled, given an arrival time in $[1, \ell]$, is $\frac{\lfloor N/e \rfloor}{\ell}$. Hence,

$$\mathbb{P}(\{a \text{ is selected}\} \mid \{a \text{ arrives at time } \ell\}) = \begin{cases} \frac{\lfloor N/e \rfloor}{\ell - 1} & \text{if } p_1 k \text{ } G_1 \text{ elements have arrived,} \\ 1 & \text{otherwise.} \end{cases}$$

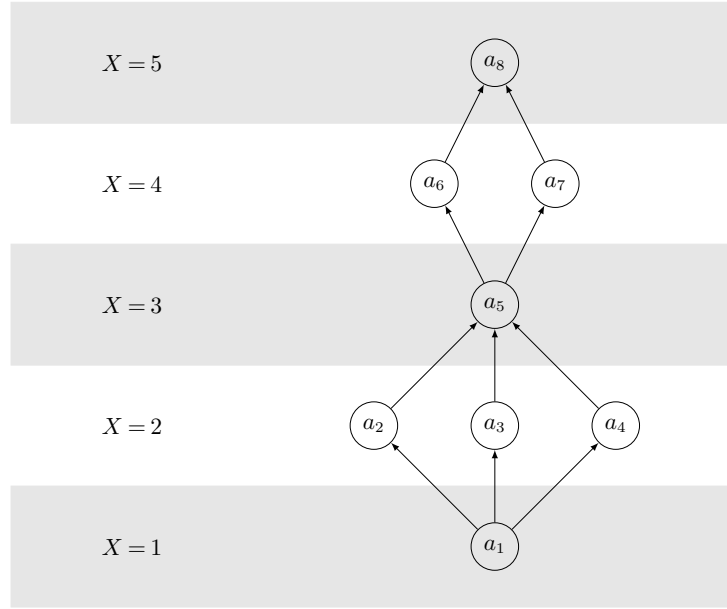


Figure 6 Partial ranking of eight elements according to the discrete attribute X . Elements in the same horizontal band have the same attribute value and are order isomorphic.

So, letting Alg denote the set of elements returned by GAPCAP , we have that

$$\begin{aligned}
 \mathbb{P}(\{a \in \text{Alg}\}) &= \sum_{\ell=\lfloor N/e \rfloor}^N \mathbb{P}(\{i \text{ arrives at time } \ell\}) \mathbb{P}(\{p_1 k \text{th best } G_1 \text{ element at time } \ell \text{ was sampled}\}) \\
 &= \frac{1}{N} \sum_{\ell=\lfloor N/e \rfloor}^N \mathbb{P}(\{p_1 k \text{th best } G_1 \text{ element at time } \ell \text{ was sampled}\}) \\
 &\geq \frac{\lfloor N/e \rfloor}{N} \sum_{\ell=\lfloor N/e \rfloor}^N \frac{1}{\ell-1} \\
 &> \frac{\lfloor N/e \rfloor}{N} \ln \left(\frac{N}{\lfloor N/e \rfloor} \right) \approx e^{-1}.
 \end{aligned}$$

It follows that $\mathbb{E}(w(\text{Alg})) \geq \frac{1}{e} w(\text{OPT}_p)$. Finally, letting OPT denote the true optimal set of elements, it remains to compare OPT_p to OPT . To that end, suppose $a \in \text{OPT}$. Then

$$\begin{aligned}
 \mathbb{P}(\{a \in \text{OPT}_p\} \mid \{a \in \text{OPT}\}) &= \sum_{j=1}^g \mathbb{P}(\{a \in G_j\}) \mathbb{P}(\{w(a) \geq p_j k \text{th best } G_j \text{ element}\} \mid \{a \in G_j\}) \\
 &= \sum_{j=1}^g p_j \mathbb{P}(\{w(a) \geq p_j k \text{th best } G_j \text{ element}\} \mid \{a \in G_j\}) \\
 &\geq \sum_{j=1}^g p_j \underbrace{\mathbb{P}(\{|G_j \cap \text{OPT}| \leq p_j k\})}_{\xrightarrow{k \rightarrow \infty} \frac{1}{2}} \\
 &= \frac{1}{2} - o(1).
 \end{aligned}$$

It follows that for any $a \in \text{OPT}$, we have

$$\mathbb{P}(\{a \in \text{Alg}\} \mid \{a \in \text{OPT}\}) \geq \left(\frac{1}{2} - o(1) \right) \mathbb{P}(\{a \in \text{Alg}\} \mid \{a \in \text{OPT}_p\}) \geq \left(\frac{1}{2} - o(1) \right) e^{-1}.$$

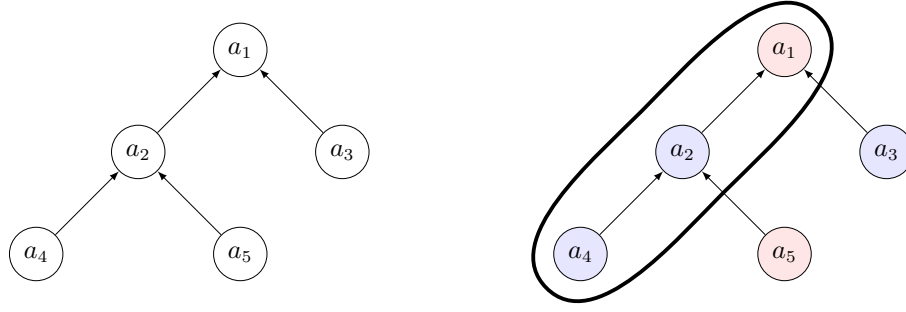


Figure 7 The input poset is shown on the left. On the right, the same poset has been annotated to show the sample S (circled) as well as the random labelling, where the blue vertices (a_2, a_3, a_4) are assigned the first label, and the red vertices (a_1, a_5) are assigned the second label.

Hence GAPCAP is $2e(1 + o(1))$ -competitive. Moreover, if $p_j < 1 - \frac{1}{k}$ for all $j \in [g]$, then $\mathbb{P}(\{|G_j \cap \text{OPT}| \leq p_j k\}) > \frac{1}{4}$ (Greenberg and Mohri 2014), thus resulting in a competitive ratio of $4e$. $\square$

Appendix D: Example Run of Algorithm 2

Consider the instance of poset bias where $k = 2$ selections can be made, and the input poset is shown on the left in Figure 7. POSETLABEL will begin by randomly choosing a sequence of k -nary labels: $(0, 0, 1, 0, 1)$. Suppose the sequence of elements that will be given to the algorithm is

$$a_2, a_4, a_1, a_3, a_5.$$

This means that $\{a_2, a_3, a_4\}$ will receive label 0, and $\{a_1, a_5\}$ will receive label 1, as shown by the coloring on the right in Figure 7.

Next, the algorithm will draw a number, say 2, from $\text{Bin}(5, 1/2)$. This will produce an initial random sample $S' = \{a_2, a_4\}$. Since the width of S' is 1, we will next draw a number, say 3, from $\text{Bin}(5, 2/3)$. The full sample S will then be the first $\max\{2, 3\} = 3$ elements, i.e., $S = \{a_1, a_2, a_4\}$.

Finally, the algorithm enters its selection phase. It first observes a_3 , which is indeed maximal in its label and thus is selected. Element a_5 is then observed but not selected, since it is ranked lower than a_1 . In this example, the set of elements returned by the algorithm would be $\{a_3\}$.

Appendix E: Algorithm for Poset Bias using Online Chain Partitioning

As discussed in Section 6, designing an algorithm for the special case of group bias is easier due to the simpler ordinal structure (recall that under group bias, the poset is comprised of a disjoint union of chains). Based on this observation, one potential approach to the poset bias setting is to sort elements into chains in an online fashion and run a competitive group bias algorithm over these chains. While there are several issues with this approach (discussed below), we propose a potential implementation.

A simple competitive algorithm requiring a width bound. Suppose we have an upper bound ω' on ω , and this is the only information we are given on the interval order. We now design an algorithm which combines our PROXYLABEL algorithm with an Online Chain Partitioning (OCP) algorithm of Kierstead and Trotter (1981). In the Online Chain Partitioning problem, elements of a partially ordered set (in this case,

Algorithm 7: GAP-CHAIN for adversarial score assignment and poset bias.

input: no. of elements N , upper bound ω' on the width, selection capacity k , OCP

 algorithm $\mathcal{A}$

```

1 Let  $g \leftarrow 3\omega' - 2$ 
2 Let  $m \sim \text{Bin}(N, \frac{g}{g+1})$  and  $S \leftarrow \{\text{first } m \text{ elements}\}$            // sample  $m$  elements
3 Assign to each  $a$  a label  $\ell(a) \in [k]$  uniformly at random
4 for  $a \in S$  do
5    $S \leftarrow S \cup \{a\}$                                            // add  $a$  to sample
6   Add  $a$  to a chain according to  $\mathcal{A}$                                // create chains  $G_1, \dots, G_g$ 
7 for  $a \notin S$  with  $\ell(a) = \ell$  do
8   Assign  $a$  to a group  $G_i$  according to  $\mathcal{A}$ 
9   if  $a$  is maximal with respect to previously seen  $\ell$ -labeled elements in  $G_i$ , and no  $\ell$ -labeled
      element has been selected then
10    Select  $a$ 

```

an interval order) are presented to an algorithm one by one, revealing any comparisons with previously seen elements. Upon arrival of an element, the algorithm must immediately and irrevocably assign the element to a chain (i.e., online greedy assignment). In Kierstead and Trotter (1981), an algorithm is given which decomposes an interval order into $3\omega - 2$ chains, where ω is the minimum number of chains possible. We can thus run this algorithm in tandem with PROXYLABEL to get a competitive algorithm (see Algorithm 7).

PROPOSITION E.1 *Suppose the underlying partial order is an interval order of width ω . Given an upper bound ω' on ω , there is a $e^2(3\omega' - 1)$ -competitive algorithm for this problem.*

Proof of Proposition E.1. Using the online chain partitioning algorithm of Kierstead and Trotter (1981), we can partition $\{a_1, \dots, a_N\}$ into $3\omega - 2$ groups in an online fashion. Given an upper bound $\omega' \geq \omega$ on ω , we know that there will be at most $3\omega' - 2$ groups. Recall that PROXYLABEL requires only an estimate of the number of groups to decide on the sampling probability for generating an initial sample. Then applying PROXYLABEL with a width proxy of $g = 3\omega' - 2$ gives an $e^2(3\omega' - 1)$ -competitive algorithm. See Algorithm 7 for the pseudocode. $\square$

Fairness of the OCP approach. One must be careful when restricting a poset to a subset of its comparisons, as that can violate ordinal fairness (see Proposition E.2 below). If an algorithm decomposes the poset into chains using only the ordinal comparisons, then ordinal fairness will still be satisfied by Proposition 3.1. However, if a chain decomposition is given to an algorithm and selections are made using that chain decomposition, ordinal fairness may fail. For instance, if the poset is the set $\{a_1, a_2, a_3\}$ with the comparisons $a_2 \prec a_1$ and $a_3 \prec a_1$, and an algorithm makes selection decisions using the chain decomposition $\{\{a_2\}, \{a_1, a_3\}\}$, then OF fails, as shown in the proof of Proposition E.2). However, the chain decomposition is generated using only ordinal information, then the decompositions $\{\{a_2\}, \{a_1, a_3\}\}$ and $\{\{a_3\}, \{a_1, a_2\}\}$ would be equally likely to be produced, and OF would not be violated.

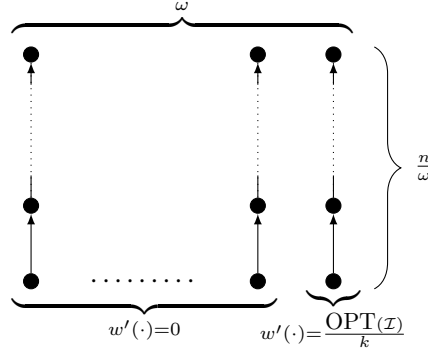


Figure 8 The poset constructed in the proof of Proposition 5.1.

PROPOSITION E.2 (CHAIN PARALLELIZATION DOES NOT SATISFY OF) *Let $\mathcal{A}$ be an algorithm for the vanilla k -secretary problem. Consider the poset secretary algorithm $\mathcal{A}_p$ which, given a chain decomposition $C_1, \dots, C_\omega$ of optimal size and a chain-membership oracle, chooses a chain $i \in [\omega]$ uniformly at random and runs $\mathcal{A}$ to select at most k elements from C_i . The algorithm $\mathcal{A}_p$ does not necessarily satisfy OF with respect to the original poset.*

Proof. Consider the algorithm $\mathcal{A}$ which selects each arriving element with probability k/n until k selections have been made. Now consider the poset $\{a_1, a_2, a_3\}$, where $a_2 \prec a_1$, $a_3 \prec a_1$, and a_2 is incomparable with a_3 , and let $k = 1$. Suppose the algorithm is given the chain decomposition $C_1 = \{a_2\}$ and $C_2 = \{a_1, a_3\}$. Then $\mathcal{A}_p$ will select a_2 with probability $1/2$ but will select a_3 with probability $1/4$. Since a_2 is order-isomorphic to a_3 , the algorithm $\mathcal{A}_p$ does not satisfy OF. $\square$

Appendix F: Missing analysis from Section 5

Proof of Proposition 5.1. Choose the number of elements $n = \min\{n : n \geq \omega k \text{ and } \omega \mid n\}$. Then choose the ordinal structure shown in Figure 8. Let a_j^i be the j th best element in the i th chain.

Let the algorithm $\mathcal{A}$ be fixed, and let p_j^i be the probability of selection of the j th best element in the i th chain. Without loss of generality, let $\omega = \arg \min_i \sum_j p_j^i$. Then $\sum_j p_j^\omega \leq k/\omega$.

We now define a scoring of the elements. First, fix some $M > 0$, which will represent the optimal total score. Let $w'(e_j^\omega) = \frac{M}{k}$ for all j , and let $w'(a_j^i) = 0$ otherwise. For the sake of generating distinct scores, we will define a perturbation w of the scoring w' . In particular, fix $\varepsilon > 0$, and let $w(\cdot)$ be a perturbation of $w'(\cdot)$ where $w(a_j^i) = w'(a_j^i) + \varepsilon_j^i$ are distinct, $0 \leq \varepsilon_j^i < \frac{M}{2k}$ for all $i < \omega$ and j , $|\varepsilon_j^\omega| < \frac{M}{2k}$ for all j , $\sum_{i < \omega, j} \varepsilon_j^i \leq \frac{\varepsilon \cdot M}{\omega}$, and $\sum_j \varepsilon_j^\omega = 0$. Then the optimum with respect to $\mathcal{I} = (\mathcal{P}, w)$ is $\text{OPT}(\mathcal{I}) = M$. So, letting $\text{Alg}(\mathcal{I})$ denote the output of the algorithm, we have that

$$\mathbb{E}[w(\text{Alg}(\mathcal{I}))] = \sum_{i < \omega, j} p_j^i \varepsilon_j^i + \sum_j p_j^\omega \left(\frac{\text{OPT}(\mathcal{I})}{k} + \varepsilon_j^\omega \right) \leq \sum_{i < \omega, j} \varepsilon_j^i + \sum_j p_j^\omega \frac{\text{OPT}(\mathcal{I})}{k} \leq \frac{(1 + \varepsilon) \text{OPT}(\mathcal{I})}{\omega}.$$

Letting $\varepsilon \rightarrow 0$, we see that $\frac{\text{OPT}(\mathcal{I})}{\mathbb{E}[w(\text{Alg}(\mathcal{I}))]} \geq \omega$. Thus, for any algorithm $\mathcal{A}$ and any fixed k and ω , we have that $\sup_{\mathcal{I}} \frac{\text{OPT}(\mathcal{I})}{\mathbb{E}[w(\text{Alg}(\mathcal{I}))]} \geq \omega$. $\square$

The following lemma and proof, which we include for completeness, is due to Kleinberg (2005).

LEMMA F.1 (KLEINBERG (2005)) Let $S = \{a_1, \dots, a_n\}$ be a set of elements, let $w : S \rightarrow [0, \infty)$ be a weighting of S with $w(a_1) \geq \dots \geq w(a_n)$, and let k be an even integer. Let Y be a random set obtained by including each element of S independently with probability $1/2$. Finally, let $OPT = \sum_{i=1}^k w(a_i)$, and let $OPT(Y)$ denote the sum of the top $k/2$ element weights in Y . Then $\mathbb{E}[OPT(Y)] \geq \left(\frac{1}{2} - \frac{1}{4\sqrt{k}}\right) OPT$.

Proof. Let $T = \{a_1, \dots, a_k\}$ be the k elements of largest weight, and let $\rho = |T \cap Y|$. Then

$$\mathbb{E}[OPT(Y) \mid \rho] \geq \begin{cases} \frac{\rho}{k} OPT & \text{if } \rho \leq \frac{k}{2} \\ \frac{1}{2} OPT & \text{else.} \end{cases}$$

Thus $\mathbb{E}[OPT(Y)] \geq \frac{\mathbb{E}[\min\{\rho, k/2\}]}{k} OPT = \frac{\mathbb{E}[\frac{1}{2}(\rho + k/2) - \frac{1}{2}|\rho - k/2|]}{k} OPT = \frac{1}{2} - \frac{\mathbb{E}[|\rho - k/2|]}{2k} OPT$. Since $\mathbb{E}[|\rho - k/2|] \leq (\mathbb{E}[(\rho - k/2)^2])^{1/2} = \sqrt{k/4}$, we have that

$$\mathbb{E}[OPT(Y)] \geq \left(\frac{1}{2} - \frac{\mathbb{E}[|\rho - k/2|]}{2k}\right) OPT \geq \left(\frac{1}{2} - \frac{1}{4\sqrt{k}}\right) OPT.$$

□

LEMMA F.2 Let $\mathcal{P}$ be a poset of width ω and $m \geq 0$ an integer. Then the number of antichains $\tau \subseteq \mathcal{P}$ with $|\sigma(\tau)| = m$ is at most $\binom{m+\omega-1}{\omega-1}$.

Proof. First, we show that there is an injection from the set of antichains to the set of selection sets. Let τ, τ' be two antichains, and suppose that $\sigma(\tau) = \sigma(\tau')$. Let a be any element in τ , and suppose for the sake of contradiction that $a \notin \tau'$. Since $a \notin \sigma(\tau)$, it must be that $a \notin \sigma(\tau')$ as well, and so $a \prec b$ for some $b \in \tau'$. Thus $b \in \sigma(\tau) \setminus \sigma(\tau')$, a contradiction.

Now let $C_1, \dots, C_\omega$ be a chain decomposition of $\mathcal{P}$, where $a_{i,1} \prec \dots \prec a_{i,n_i}$ are the elements of C_i . Define $C_{i,k} = \{a_{i,1}, \dots, a_{i,k}\}$ to be the prefix of C_i of length k .

For any τ with $|\sigma(\tau)| = m$, we must have that $\sigma(\tau) = \bigcup_{i=1}^\omega C_{i,x_i}$, for some $x_1 + \dots + x_\omega = m$. The number of such sets $\sigma(\tau)$ is bounded by the number of nonnegative integral solutions to $x_1 + \dots + x_\omega = m$, which is $\binom{m+\omega-1}{\omega-1}$. Moreover, by the above injection, the number of antichains τ with $|\sigma(\tau)| = m$ is bounded by the same. □

LEMMA F.3 Let $\mathcal{P}$ be a fixed poset on N elements of width ω , and let $\hat{\tau}$ be the threshold chosen by Algorithm 3 on input of $\eta \in (0, 1]$ and $k \in \mathbb{N}$. Then $B \leq Ne^{-\eta^2 k/4\omega}$, where $B = \mathbb{P}\left(\bigcup_{b \in \hat{\tau}} \left\{|\mathcal{U}(b)| < (1-2\eta)\frac{k}{\omega}\right\}\right)$.

Proof.

$$\begin{aligned} B &= \mathbb{P}\left(|\mathcal{U}(b)| < (1-2\eta)\frac{k}{\omega} \text{ for some } b \in \hat{\tau}\right) \\ &\leq \sum_{b: |\mathcal{U}(b)| \leq (1-2\eta)\frac{k}{\omega}} \mathbb{P}(b \in \hat{\tau}) \\ &\leq \sum_{b: |\mathcal{U}(b)| \leq (1-2\eta)\frac{k}{\omega}} \mathbb{P}\left(\underbrace{|\mathcal{U}(b) \cap Y|}_{\sim \text{Bin}(\lfloor (1-2\eta)k/\omega \rfloor, 1/2)} \geq (1-\eta)\frac{k}{2\omega}\right) \\ &\leq \sum_{b: |\mathcal{U}(b)| \leq (1-2\eta)\frac{k}{\omega}} \mathbb{P}\left(|\mathcal{U}(b) \cap Y| - \frac{1}{2}\left\lfloor (1-2\eta)\frac{k}{\omega} \right\rfloor \geq \frac{\eta k}{2\omega}\right) \\ &\stackrel{(c)}{\leq} Ne^{-\eta^2 k/2(1-2\eta)\omega} \\ &\leq Ne^{-\eta^2 k/4\omega}, \end{aligned}$$

where (c) follows from Hoeffding's inequality. □

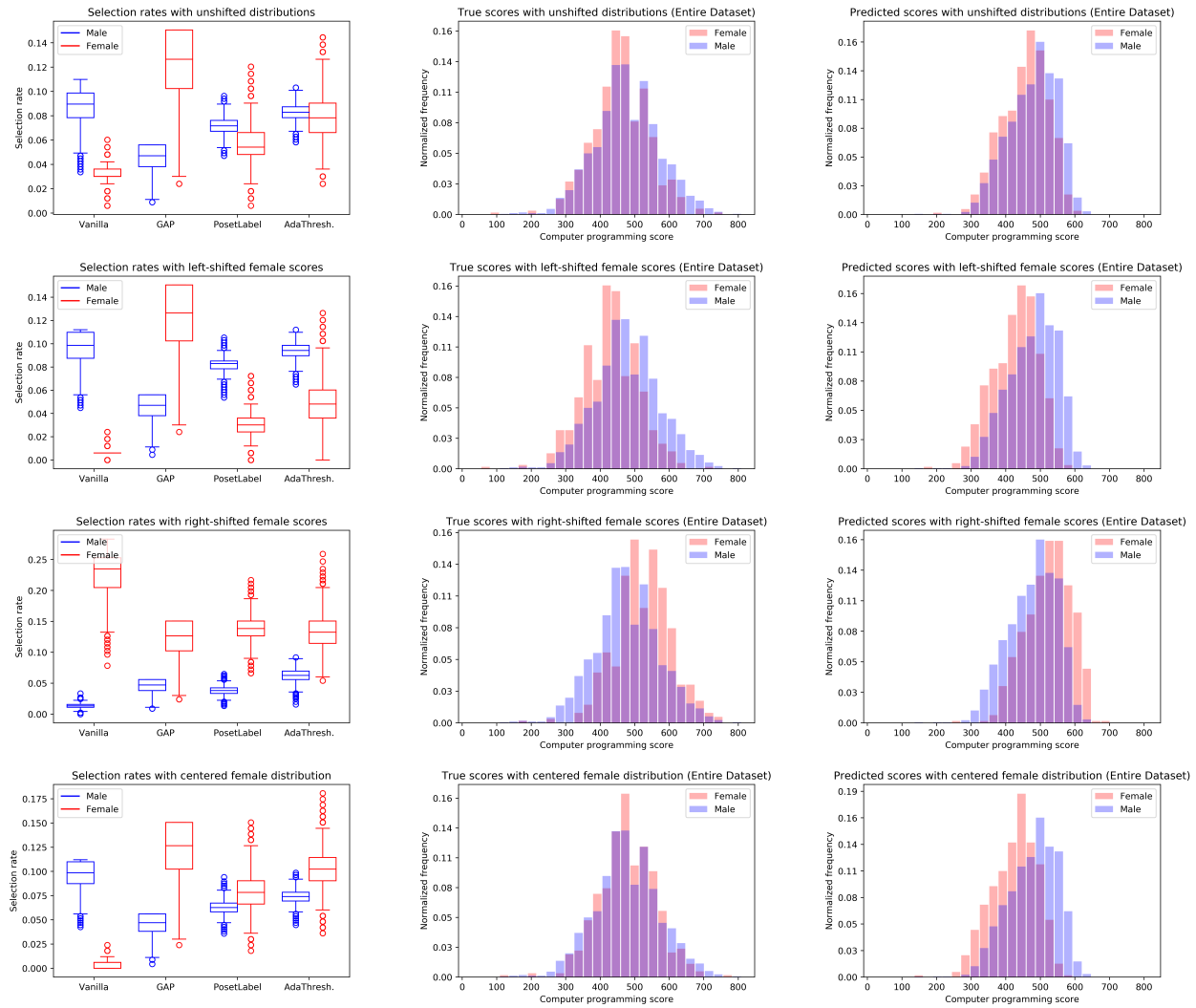


Figure 9 Experiment results for selection capacity $k = 50$ and parameter $\lambda = 1/6$. The four algorithms are run on unshifted scores (top), left-shifted female scores (second row), right-shifted female scores (third row), and centered data (i.e., the true scores of the female group were translated so that both groups have the same mean true score) with a prediction model modified to be biased (bottom). Shown above are the resulting selection rates (left), the true score distributions (center), and the predicted score distributions (right).

Appendix G: Additional experiments

In this section, we perform the same experiments as in Section 7 but with several modifications: we consider a selection capacity of $k = 50$ with the same confidence parameter $\lambda = 1/6$ in Figure 9, we consider $k = 25$ with $\lambda = 1/12$ in Figure 10, and we consider $k = 25$ with $\lambda = 1/3$ in Figure 11. In general, we see the same trends seen in the experiments of Section 7: the poset-based algorithms are sensitive to the true score distributions, but do not blindly match proportional selection rates as the quota-based GAP. Moreover, by comparing model-predicted scores to the training data, the poset-based algorithms effectively counteract bias introduced by the prediction model.

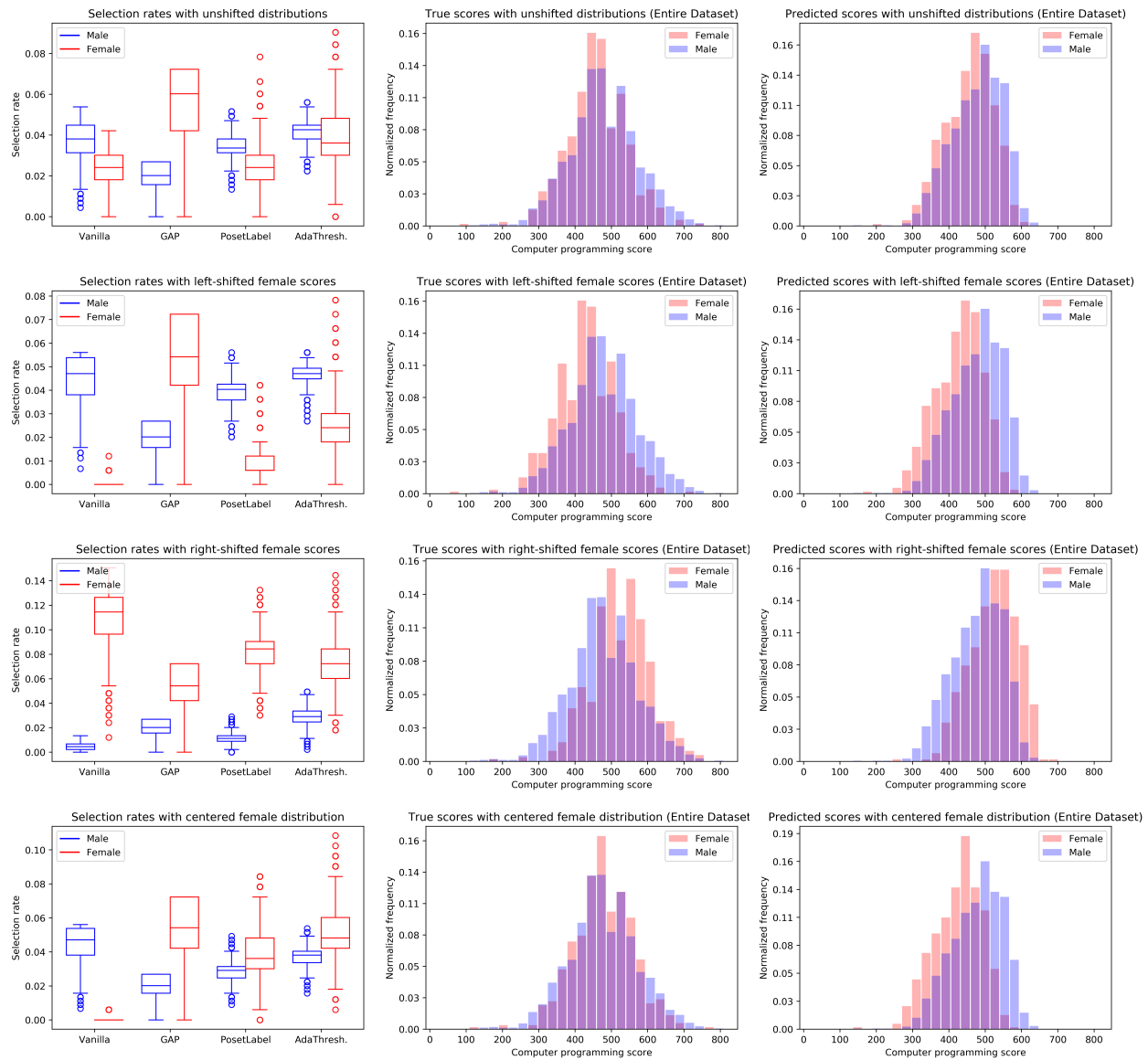


Figure 10 Experiment results for selection capacity $k = 25$ and parameter $\lambda = 1/12$. The four algorithms are run on unshifted scores (top), left-shifted female scores (second row), right-shifted female scores (third row), and centered data (i.e., the true scores of the female group were translated so that both groups have the same mean true score) with a prediction model modified to be biased (bottom). Shown above are the resulting selection rates (left), the true score distributions (center), and the predicted score distributions (right).

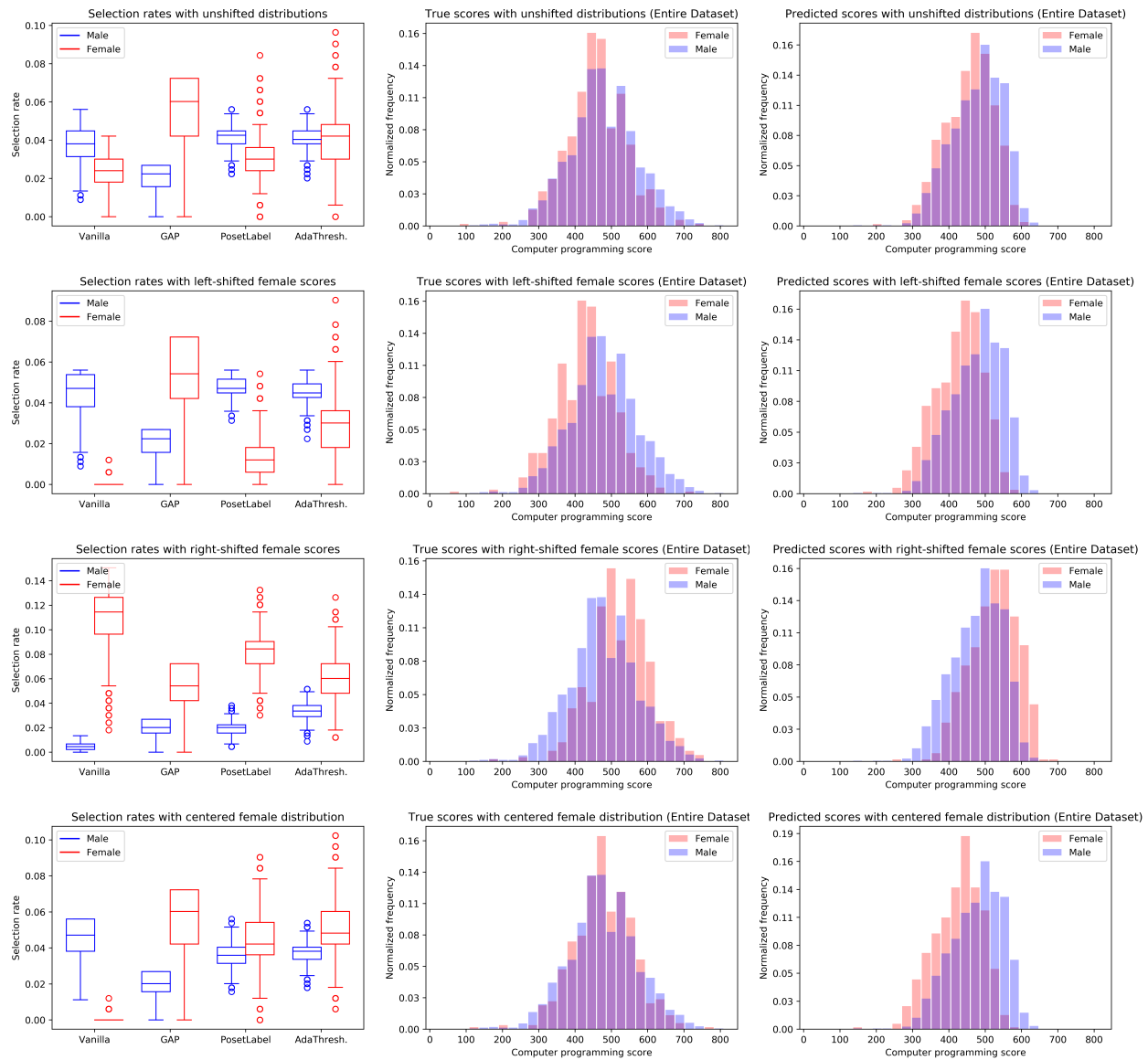


Figure 11 Experiment results for selection capacity $k = 25$ and parameter $\lambda = 1/3$. The four algorithms are run on unshifted scores (top), left-shifted female scores (second row), right-shifted female scores (third row), and centered data (i.e., the true scores of the female group were translated so that both groups have the same mean true score) with a prediction model modified to be biased (bottom). Shown above are the resulting selection rates (left), the true score distributions (center), and the predicted score distributions (right).

Gender	10percent	12percent	College tier	College GPA	College city tier	English	Logical
-16.95	-0.2193	0.2372	-18.50	1.182	1.563	0.02541	0.1429
Quant	Domain	ElectronicsAndSemicon		Computer science	Mechanical eng.	Electrical eng.	
0.1199	177.5	-0.09960		0.006473	-0.3314	-80.72	
Telecom. eng.		Civil eng.	Conscientiousness	Agreeableness	Extraversion	Neuroticism	
-80.72		0.4119	-4.598	2.649	-3.256	-4.508	
Openness to experience				Graduation age			
3.565				0.1764			

Table 2 Coefficients (partial derivatives) of the linear prediction model for computer programming, trained on the unshifted data.